CareTrack

Hospital Operations & Corporate Wellness Management System

Built on Microsoft Power Platform (Dataverse, Model-Driven App, Canvas App, Power Automate, Copilot Studio)

Project Report

Digital Transformation: Tools and Techniques

[SE4010 / BCS-8A]

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1. Executive Summary

CareTrack is a hospital operations and corporate wellness management system built entirely on Microsoft's Power Platform. It addresses a real and growing business need in healthcare delivery: hospitals increasingly partner with employers to deliver healthcare benefits to staff under contracted wellness packages, in addition to serving walk-in patients directly. CareTrack digitizes this entire relationship — from patient registration and appointment booking, through doctor scheduling and clinical record-keeping, to corporate contract management and billing — within a single, integrated Dataverse-backed solution.

The system is composed of five integrated layers: a normalized Dataverse data model of nine interrelated tables; a Model-Driven App used by hospital staff (administrators, doctors, nurses, receptionists, finance, and pharmacists) with role-based security; a Canvas App patient portal used by patients and corporate HR contacts to register, look up their identity, book appointments, and manage corporate wellness enquiries; a library of Power Automate flows that automate notifications, record creation, and business rule enforcement; and a Copilot Studio conversational agent that allows patients to book appointments through natural conversation rather than navigating a form.

Beyond basic CRUD functionality, CareTrack demonstrates several deliberate design decisions that reflect real-world system design: a Business-to-Business-to-Consumer (B2B2C) billing model with contract-based corporate accounts, a Business Process Flow visualizing the corporate sales pipeline, role-based and field-level (column) security, scenario-driven Assign/Share permissions, and a layered automation strategy spanning more than twenty Power Automate flows covering everything from appointment confirmations to contract renewal reminders.

2. Project Idea

2.1 Problem Statement

Healthcare access in many organizations remains fragmented across three disconnected groups: hospitals that manage doctors, schedules, and clinical records largely on paper or in siloed legacy systems; corporate HR departments that purchase health benefits for employees through manual, email-driven negotiation processes with little ongoing visibility into utilization; and patients who must navigate phone calls, walk-in queues, and inconsistent record-keeping to access care. There is no single platform that connects all three parties — hospital, employer, and patient — into one coherent digital experience.

2.2 Proposed Solution

CareTrack proposes a unified Power Platform solution that serves all three stakeholders from the same underlying data model:

- Hospital staff manage departments, doctors, schedules, appointments, clinical records, and billing through a role-secured Model-Driven App.
- Corporate HR teams submit wellness package enquiries, negotiate terms, and track their contract lifecycle — visualized through a dedicated Business Process Flow.
- Patients — whether walk-in or covered under a corporate package — register, look themselves up by phone number (with multi-profile support for families), book appointments against real doctor availability, and track their appointment and billing history through a Canvas App patient portal.
- A conversational Copilot Studio agent gives patients an alternative, chat-driven booking channel that mirrors a human receptionist.
- Power Automate flows tie the system together end to end — confirming bookings, generating medical records and invoices automatically on appointment completion, alerting staff to no-shows, and reminding corporate clients of upcoming contract renewals.

2.3 Evolution from Initial Proposal

The project's working scope evolved from an initial “clinic” framing into a multi-department “hospital” framing during design. This was a deliberate refinement rather than a deviation: the B2B2C corporate wellness model only becomes meaningful at a scale where there are multiple departments and specialists across which package tiers (Basic, Standard, Premium) can differentiate access. A single-doctor clinic has nothing to meaningfully tier. Critically, this shift in branding and scope required almost no rework of the underlying Dataverse model — the same nine tables and relationships support both framings, with only the addition of a Hospital Department table and a small number of additional doctor records. This demonstrates that the data architecture was designed to scale from the outset.

3. Business Value

3.1 Why a B2B2C Model

Rather than building a generic patient-booking application — the default approach for most student projects in this space — CareTrack models a genuine B2B2C (Business-to-Business-to-Consumer) relationship, reflecting how corporate wellness partnerships actually function in the healthcare industry (a pattern used by providers such as Cigna, Allianz Care, and regionally by services such as Sehat Kahani). A hospital (the business) contracts with employers (the other business) to deliver healthcare benefits to employees (the consumers), with the hospital managing contract value, payment frequency, package tiers, and per-employee visit allowances on an ongoing basis.

This model introduces genuine business complexity worth digitizing: tiered package access, per-employee visit caps that reset monthly, contract-based billing independent of individual visit volume, and three distinct billing pathways (walk-in, corporate, and insurance) that converge on

a single Invoice table with conditional logic determining how much of each visit's cost is covered versus payable by the patient.

3.2 Value Delivered to Each Stakeholder

Stakeholder	Value Delivered
Hospital Administration	Centralized visibility across departments, doctors, schedules, patients, and billing; role-based access ensures staff only see what their job requires; automated record creation reduces manual data entry after every appointment.
Corporate HR / Employers	Self-service enquiry submission, a visualized negotiation-to-activation pipeline, automated renewal reminders, and ongoing visibility into employee benefit utilization — replacing ad hoc email-based account management.
Patients / Employees	Self-service registration and booking without phone queues; support for multiple family profiles under one contact number; real-time doctor availability based on actual working hours; transparent billing showing exactly what is covered versus payable.
Doctors	A schedule-driven booking system that prevents double-booking; the ability to reassign appointments for referrals and share records for second opinions without losing ownership of their own data.

3.3 Alignment with Digital Transformation Theme

The course's central theme — digital transformation tools and techniques — is best demonstrated not by digitizing a single transaction (e.g., “book an appointment”) but by digitizing an entire business relationship and the operational rules that govern it. CareTrack's contract management, tiered billing, role-based security, automated notification pipeline, and conversational AI booking channel collectively demonstrate a system designed around real operational processes — not a CRUD demo wrapped in a healthcare theme.

4. Entity Relationship Diagram (ERD)

4.1 Data Model Overview

CareTrack's Dataverse data model consists of nine interrelated tables, organized around three logical clusters: clinical operations (Hospital Department, Doctor, Doctor Schedule, Appointment, Medical Record), corporate wellness administration (Corporate Account, Employee), and patient/billing records (Patient, Invoice). Each table follows standard normalization principles (1NF/2NF/3NF) with a small number of deliberately denormalized lookups, justified in the end of Section 4.3.

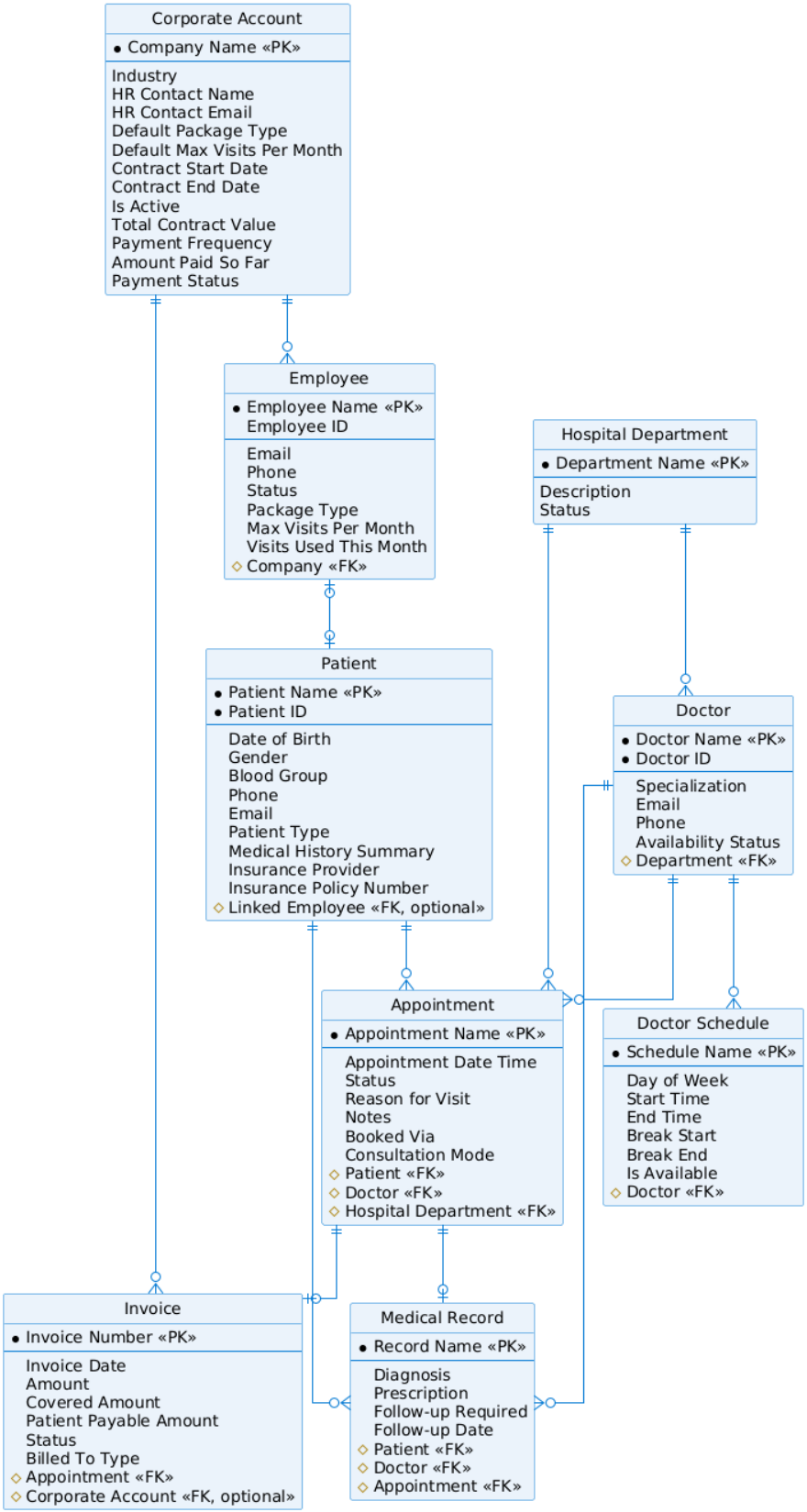

Corporate Account
• Company Name «PK»
Industry
HR Contact Name
HR Contact Email
Default Package Type
Default Max Visits Per Month
Contract Start Date
Contract End Date
Is Active
Total Contract Value
Payment Frequency
Amount Paid So Far
Payment Status
Employee
• Employee Name «PK»
Employee ID
Email
Phone
Status
Package Type
Max Visits Per Month
Visits Used This Month
◇ Company «FK»
Hospital Department
• Department Name «PK»
Description
Status
Patient
• Patient Name «PK»
• Patient ID
Date of Birth
Gender
Blood Group
Phone
Email
Patient Type
Medical History Summary
Insurance Provider
Insurance Policy Number
◇ Linked Employee «FK, optional»
Doctor
• Doctor Name «PK»
• Doctor ID
Specialization
Email
Phone
Availability Status
◇ Department «FK»
Appointment
• Appointment Name «PK»
Appointment Date Time
Status
Reason for Visit
Notes
Booked Via
Consultation Mode
◇ Patient «FK»
◇ Doctor «FK»
◇ Hospital Department «FK»
Doctor Schedule
• Schedule Name «PK»
Day of Week
Start Time
End Time
Break Start
Break End
Is Available
◇ Doctor «FK»
Invoice
• Invoice Number «PK»
Invoice Date
Amount
Covered Amount
Patient Payable Amount
Status
Billed To Type
◇ Appointment «FK»
◇ Corporate Account «FK, optional»
Medical Record
• Record Name «PK»
Diagnosis
Prescription
Follow-up Required
Follow-up Date
◇ Patient «FK»
◇ Doctor «FK»
◇ Appointment «FK»

Fig 1. ERD

4.2 Entity Definitions

4.2.1 Hospital Department

Field	Type	Required
Department Name (Primary Key)	Text	Yes
Description	Multiline Text	No
Status	Choice (Active, Inactive)	No

4.2.2 Doctor

Field	Type	Required
Doctor Name (Primary Key)	Text	Yes
Doctor ID	Text	Yes
Specialization	Text	No
Email	Email	No
Phone	Phone	No
Availability Status	Choice (Available, On Leave, Busy)	No
Department (Foreign Key)	Lookup → Hospital Department	Yes

4.2.3 Doctor Schedule

Field	Type	Required
Schedule Name (Primary Key)	Text (auto-generated)	Yes
Day of Week	Choice (Monday–Sunday)	No
Start Time	Text (24-hour format)	No
End Time	Text (24-hour format)	No
Break Start	Text (24-hour format)	No
Break End	Text (24-hour format)	No
Is Available	Yes/No	No
Doctor (Foreign Key)	Lookup → Doctor	Yes

Note: Start/End/Break times are stored as text rather than a native Time type because Dataverse does not offer a time-only field type independent of date — only "Date and Time" or "Date Only" are available. Storing as 24-hour text avoids unnecessary date-component storage and is parsed at the application layer for schedule and slot-availability calculations.

4.2.4 Corporate Account

Field	Type	Required
Company Name (Primary Key)	Text	Yes

Field	Type	Required
Industry	Text	No
HR Contact Name	Text	No
HR Contact Email	Email	No
HR Contact Phone	Phone	No
Default Package Type	Choice (Basic, Standard, Premium)	No
Default Max Visits Per Month	Whole Number	No
Contract Start Date	Date	No
Contract End Date	Date	No
Is Active	Yes/No	No
Number of Enrolled Employees	Whole Number	No
Total Contract Value	Currency	Yes
Payment Frequency	Choice (One-time, Monthly, Quarterly, Annually)	Yes
Amount Paid So Far	Currency	No
Payment Status	Choice (Pending, Partial, Paid)	No

4.2.5 Employee

Field	Type	Required
Employee Name (Primary Key)	Text	Yes
Employee ID	Text	No
Email	Email	No
Phone	Phone	No
Status	Choice (Active, Inactive)	No
Package Type (override)	Choice (Basic, Standard, Premium)	No
Max Visits Per Month (override)	Whole Number	No
Visits Used This Month	Whole Number (defaults to 0)	No
Company (Foreign Key)	Lookup → Corporate Account	Yes

Package Type and Max Visits Per Month follow a default-and-override pattern: when left blank, an employee inherits the company-level defaults from their Corporate Account; only exception employees (e.g., senior staff on an upgraded tier) require an explicit override. This minimizes data entry for large employee counts.

4.2.6 Patient

Field	Type	Required
Patient Name (Primary Key)	Text	Yes
Patient ID	Text	Yes
Date of Birth	Date	No
Gender	Choice (Male, Female, Other)	No
Blood Group	Choice (A+, A-, B+, B-, O+, O-, AB+, AB-, Don't Know)	No
Phone	Phone	No
Email	Email	No
Patient Type	Choice (Walk-in, Corporate)	No
Medical History Summary	Multiline Text	No
Insurance Provider	Text	No
Insurance Policy Number	Text	No
Linked Employee (Foreign Key)	Lookup → Employee (optional)	No

A single phone number may be linked to multiple Patient records — supporting both family members sharing one contact number, and the same individual holding separate Walk-in and Corporate records for different billing contexts. See Section 7 (Canvas App) for how the patient portal surfaces this through a profile-selection screen.

4.2.7 Appointment

Field	Type	Required
Appointment Name (Primary Key)	Text (auto-generated)	Yes
Appointment Date Time	Date and Time	No
Status	Choice (Scheduled, Completed, Cancelled, No Show)	No
Reason for Visit	Multiline Text	No
Notes	Multiline Text	No
Booked Via	Choice (Walk-in, Canvas App, Voice Agent,Agent)	No
Consultation Mode	Choice (In-Person, Teleconsultation)	No
Patient (Foreign Key)	Lookup → Patient	Yes
Doctor (Foreign Key)	Lookup → Doctor	Yes
Hospital Department (Foreign Key)	Lookup → Hospital Department	Yes

4.2.8 Medical Record

Field	Type	Required
Record Name (Primary Key)	Text (auto-generated)	Yes

Field	Type	Required
Diagnosis	Multiline Text	No
Prescription	Multiline Text	No
Follow-up Required	Yes/No	No
Follow-up Date	Date	No
Patient (Foreign Key)	Lookup → Patient	Yes
Doctor (Foreign Key)	Lookup → Doctor	No
Appointment (Foreign Key)	Lookup → Appointment	Yes

4.2.9 Invoice

Field	Type	Required
Invoice Number (Primary Key)	Text (auto-generated)	Yes
Invoice Date	Date	No
Amount / Total Amount	Currency	No
Covered Amount	Currency	No
Patient Payable Amount	Currency	No
Status	Choice (Pending, Paid)	No
Billed To Type	Choice (Walk-in, Corporate, Insurance)	No
Appointment (Foreign Key)	Lookup → Appointment	Yes
Corporate Account (Foreign Key)	Lookup → Corporate Account (optional)	No

4.3 Relationships Summary

Parent Table	Cardinality	Child Table
Hospital Department	1 : Many	Doctor
Doctor	1 : Many	Doctor Schedule
Doctor	1 : Many	Appointment
Hospital Department	1 : Many	Appointment
Corporate Account	1 : Many	Employee
Employee	1 : 0..1	Patient (optional, corporate patients only)
Patient	1 : Many	Appointment
Appointment	1 : 0..1	Medical Record
Appointment	1 : 0..1	Invoice
Patient	1 : Many	Medical Record
Doctor	1 : Many	Medical Record

Parent Table	Cardinality	Child Table
Corporate Account	1 : Many	Invoice (only when Billed To Type = Corporate)

Note on intentional denormalization: A small number of lookups (e.g., Appointment → Hospital Department, which is technically derivable via Appointment → Doctor → Department) are kept as direct links rather than relying on multi-hop joins. This is a deliberate, controlled denormalization — standard practice in production CRM/ERP systems — used to improve reporting performance (e.g., "appointments by department" views without a two-hop join) and to preserve historical accuracy (e.g., if a doctor later changes department, the appointment record still reflects the department at the time of the visit).

5. Security Architecture

5.1 Overview

CareTrack implements defense-in-depth security across two layers: table-level Role-Based Access Control (RBAC) governing which tables each staff role can access and what operations they can perform, and field-level Column Security Profiles that further restrict specific sensitive fields even within tables a role can otherwise access. Six security roles model realistic hospital staff functions, and three column security profiles mask sensitive fields (insurance data, clinical history, and diagnosis details) from roles that do not require them.

5.2 Security Roles Overview

Role	Description
Admin	Full organization-wide access to all tables (mapped to the built-in System Administrator role).
Doctor	Manages own appointments, schedules, and medical records for their patients.
Nurse	Assists with scheduling, patient records, and appointments; supports doctors.
Receptionist	Front-desk operations — patient registration and appointment booking.
Finance	Manages billing, invoices, and corporate account contracts.
Pharmacist	Views prescriptions from medical records to dispense medication.

5.3 Table-Level Permission Matrix

Access levels: None (no visibility), Read (view only), Read/Write (view, create, edit), Read/Write (own) (limited to records the user is directly linked to — e.g., a doctor sees only their own appointments), Full (organization-wide, Admin only).

Table	Admin	Doctor	Nurse	Reception.	Finance	Pharm.
Hospital Department	Full	Read	Read	Read	Read	Read
Doctor	Full	Read (own)	Read	Read	Read	None
Doctor Schedule	Full	R/W (own)	R/W	Read	None	None
Corporate Account	Full	None	None	Read	R/W	None
Employee	Full	None	None	Read	Read	None
Patient	Full	Read	R/W	R/W	Read	None
Appointment	Full	R/W (own)	R/W	R/W	Read	None
Medical Record	Full	R/W (own)	Read	None	None	Read*
Invoice	Full	None	None	None	R/W	None

** Pharmacist's Medical Record access is intentionally narrow — see Section 5.5 for how Column Security restricts this further to Prescription only, masking Diagnosis.*

5.4 Assign and Share — Scenario-Driven, Not Blanket

Assign (transferring record ownership) and Share (granting access without changing ownership) are left disabled across nearly all role/table combinations — deliberately, because most combinations have no realistic business scenario behind them (e.g., there is no real-world case for a Receptionist “reassigning” a Corporate Account). Two targeted exceptions were added to the Doctor role, reflecting genuine clinical workflows:

Table	Permission	Real-World Scenario
Appointment	Assign = User	A doctor can reassign their appointment to another doctor — for referrals or when unavailable.
Medical Record	Share = User	A doctor can share a patient's medical record with another doctor for a second opinion, without transferring ownership.

This demonstrates deliberate, scenario-driven security design rather than uniform default configuration applied without justification.

5.5 Column Security Profiles — Field-Level Masking

Beyond table-level access, three Column Security Profiles restrict visibility of specific sensitive fields, even for roles that otherwise have access to the containing table. Column security in Dataverse operates on an allow-list basis: once enabled on a field, that field is hidden from everyone by default unless they belong to a profile explicitly granting access.

Profile	Secured Field	Table	Visible To	Hidden From
Insurance Data Access	Insurance Policy Number	Patient	Admin, Finance	Doctor, Nurse, Receptionist, Pharmacist
Clinical History Access	Medical History Summary	Patient	Admin, Doctor, Nurse	Receptionist
Diagnosis Access	Diagnosis	Medical Record	Admin, Doctor, Nurse	Pharmacist

The Diagnosis Access profile is what makes the Pharmacist role's restriction meaningful: Pharmacist holds table-level Read access to Medical Record specifically in order to view Prescription details for dispensing, but without column-level masking they would also see the clinical Diagnosis — information beyond what their role requires. This profile narrows their effective visibility down to exactly what the job needs.

5.6 Setup Process Summary

1. Five custom security roles (Doctor, Nurse, Receptionist, Finance, Pharmacist) were created by copying an existing baseline role and customizing table-level permissions per the matrix in Section 5.3; Admin uses the built-in System Administrator role.
2. Column-level security was enabled on three fields (Insurance Policy Number, Medical History Summary, Diagnosis) via each field's Advanced Options.

3. Three Column Security Profiles were created in the CareTrack solution, each scoped to its secured field with Read/Update/Create permissions, and populated with the appropriate member users.

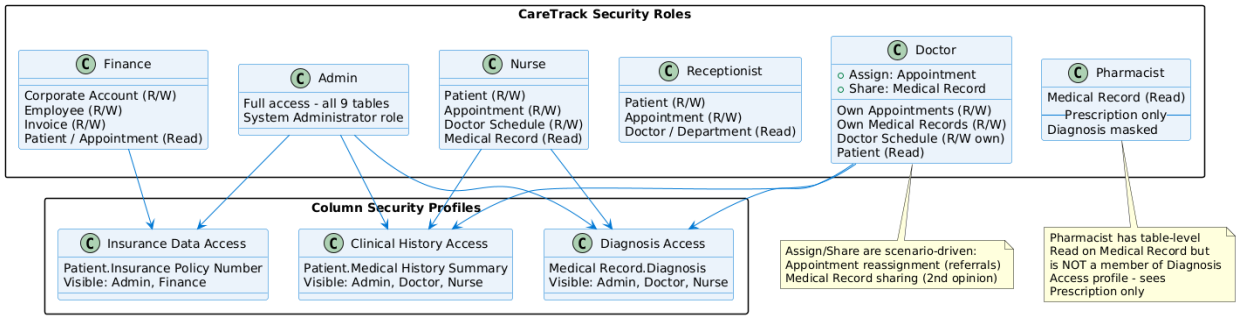


Fig. Security Diagram

CareTrack > **Security roles**

Display name ↑	Name	Type	Managed	Customized	Last Modif...
Doctor	Doctor	Security Role	No	Yes	4 days ago
Finance	Finance	Security Role	No	Yes	4 days ago
Nurse	Nurse	Security Role	No	Yes	4 days ago
Pharmacist	Pharmacist	Security Role	No	Yes	4 days ago
Receptionist	Receptionist	Security Role	No	Yes	4 days ago

Fig 2. Users

Name	Record owner...	Permission S...	Create	Read	Write	Delete	Append	Append to	Assign	Share
new_appointment	User or Team	Custom	User	User	User	User	User	User	User	None
new_doctor	User or Team	Custom	None	User	None	None	None	None	None	None
new_doctorschedule	User or Team	Custom	User	User	User	User	User	User	None	None
new_hospitaldepartment	User or Team	Reference	None	Organization	None	None	None	None	None	None
new_medicalrecord	User or Team	Custom	User	User	User	User	User	User	None	User
new_patient	User or Team	Reference	None	Organization	None	None	None	None	None	None

Fig 3. Doctor Role Access

Table ↑		Name	Record owner...	Permission S...	Create	Read	Write	Delete	Append	Append to	Assign	Share
Appointment	...	new_appointment	User or Team	Reference	None	Organization	None	None	None	None	None	None
Corporate Account	...	new_corporateaccount	User or Team	Custom	User	User	User	User	User	User	None	None
Doctor	...	new_doctor	User or Team	Reference	None	Organization	None	None	None	None	None	None
Employee	...	new_employee	User or Team	Custom	User	User	User	None	User	User	None	None
Hospital Department	...	new_hospitaldepartment	User or Team	Reference	None	Organization	None	None	None	None	None	None
Invoice	...	new_invoice	User or Team	Custom	User	User	User	User	User	User	None	None
Patient	...	new_patient	User or Team	Reference	None	Organization	None	None	None	None	None	None

Fig 4. Finance Role Access

Doctor	...	new_doctor	User or Team	Reference	None	Organization	None	None	None	None	None	
Doctor Schedule	...	new_doctorschedule	User or Team	Custom	User	User	User	User	User	User	None	
Hospital Department	...	new_hospitaldepartment	User or Team	Reference	None	Organization	None	None	None	None	None	
Medical Record	...	new_medicalrecord	User or Team	Custom	None	User	None	None	None	None	None	
Patient	...	new_patient	User or Team	Custom	User	User	User	None	User	User	None	

Fig 5. Nurse Role Access

Medical Record	...	new_medicalrecord	User or Team	Custom	None	User	None	None	None	None	None	N

Fig 6. Pharmacist Role Access

Table ↑		Name	Record ownership	Permission Settin...	Create	Read	Write	Delete	Append
Custom Tables (5)									
Appointment	...	new_appointment	User or Team	Custom	User	User	User	User	User
Doctor	...	new_doctor	User or Team	Reference	None	Organization	None	None	None
Doctor Schedule	...	new_doctorschedule	User or Team	Reference	None	Organization	None	None	None
Hospital Department	...	new_hospitaldepartment	User or Team	Reference	None	Organization	None	None	None
Patient	...	new_patient	User or Team	Custom	User	User	User	None	User

Fig 7. Receptionist Role Access

CareTrack > **Column security profiles**

	Display name ↑		Name	Description	Type	Managed
	Clinical History Access	⋮	Clinical History A...	Grants access to view and edit the Medical History Summary field on Patient recor...	Column Security ...	No
	Diagnosis Access	⋮	Diagnosis Access	Grants access to view and edit the Diagnosis field on Medical Records.	Column Security ...	No
	Insurance Data Access	⋮	Insurance Data A...	Grants access to view and edit the Insurance Policy Number field on Patient records.	Column Security ...	No

Fig 8. Column Level Security

5.7 Testing Limitation

All security roles and column security profiles were fully configured as documented above; however, live verification of role-based masking could not be visually demonstrated in this submission, since development was performed with a single shared Administrator-level test account, which bypasses both table-level restrictions and column-level masking by design. The configurations themselves — role permission matrices and column security profile membership — are captured via the screenshots above as evidence of correct setup.

6. Model-Driven App — CareTrack Admin Portal

6.1 Purpose and Audience

The CareTrack Admin Portal is the internal-facing Model-Driven App used by hospital staff — administrators, doctors, nurses, receptionists, finance staff, and pharmacists — to manage the full operational dataset. All nine Dataverse tables are exposed through a single navigation group, with each user's visible tables and permitted actions governed by the security roles described in Section 5.

6.2 Navigation and Structure

- Single sitemap group ("CareTrack") containing all nine tables: Hospital Department, Doctor, Doctor Schedule, Corporate Account, Employee, Patient, Appointment, Medical Record, and Invoice.
- Standard forms for each table reflecting the field structure documented in Section 4.2.
- Custom views configured for common staff workflows (e.g., Today's Scheduled Appointments, Pending Invoices, Active Corporate Accounts).
- The Corporate Account form additionally surfaces the Business Process Flow described in Section 6.3, guiding staff through the corporate sales pipeline stage by stage.

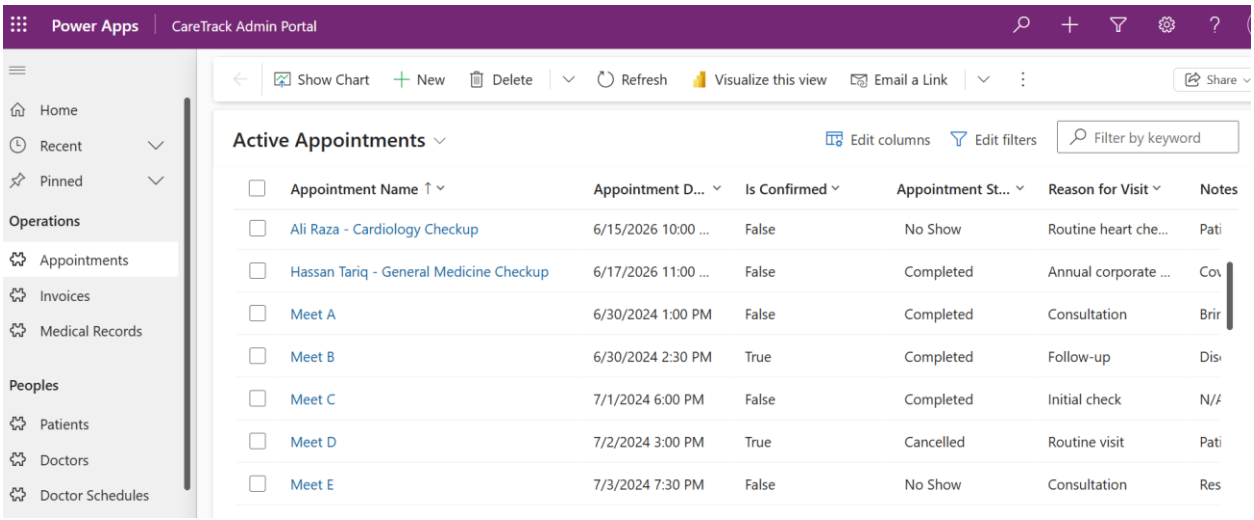


Fig 9. Model-Driven App (Admin Portal)

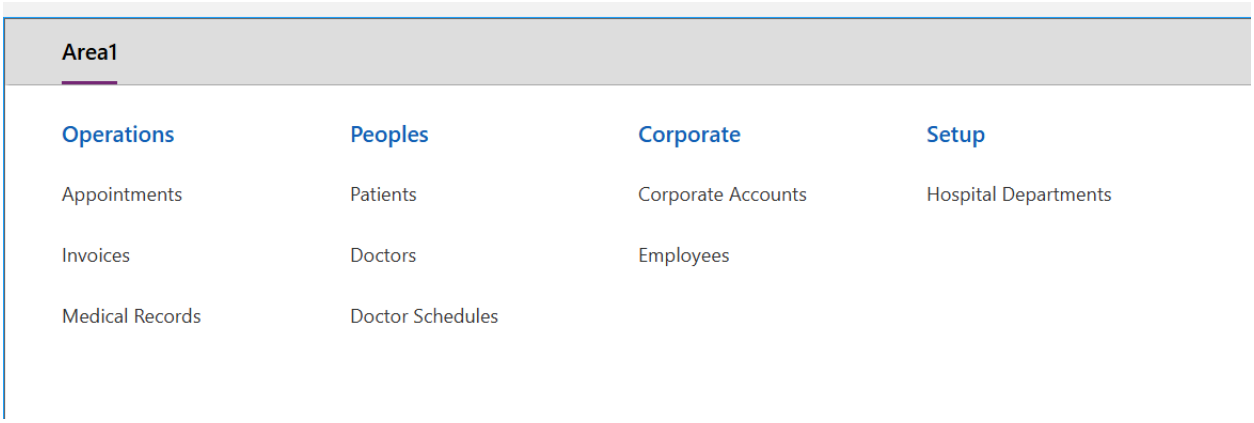


Fig 10. Sitemap for Admin Panel

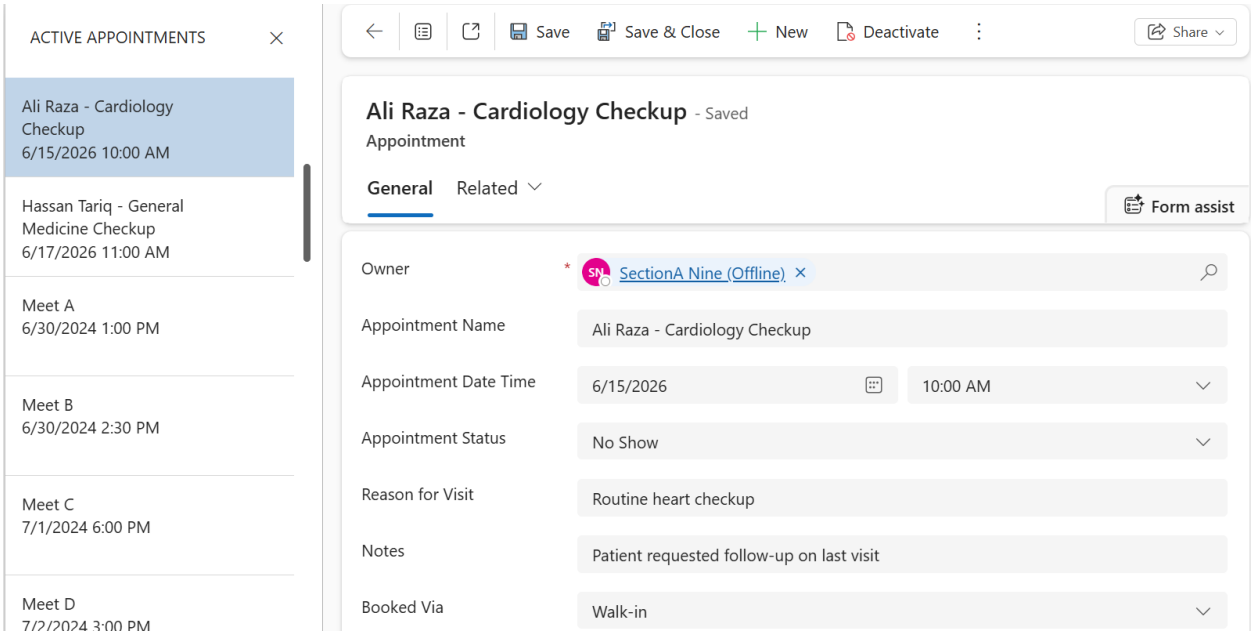


Fig 11. Appointments Form

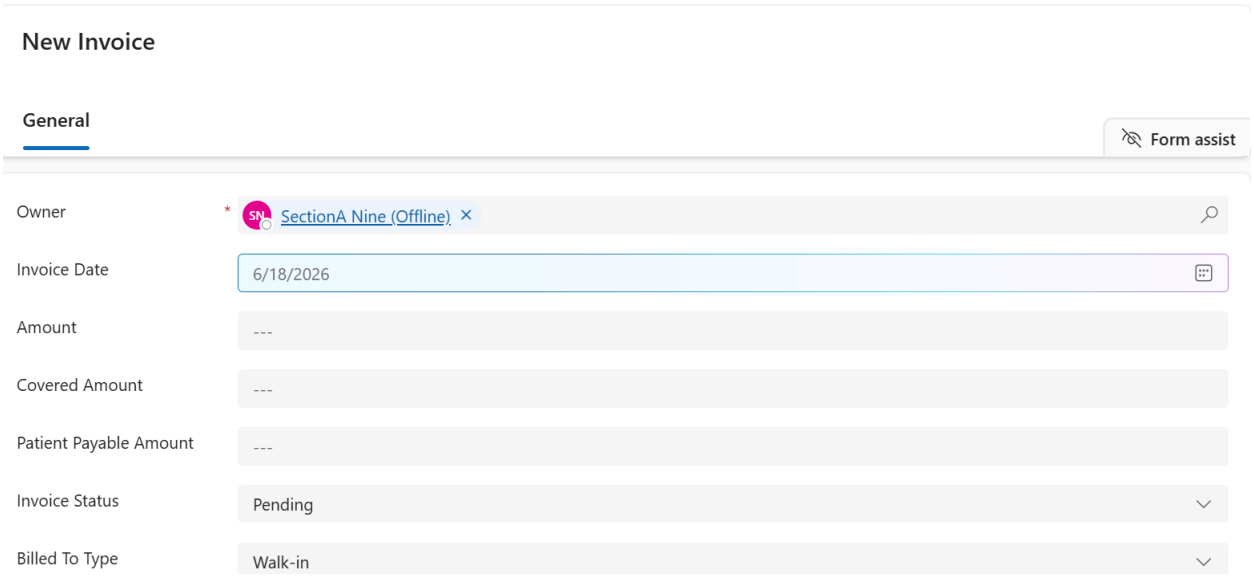


Fig 12. Invoice Form

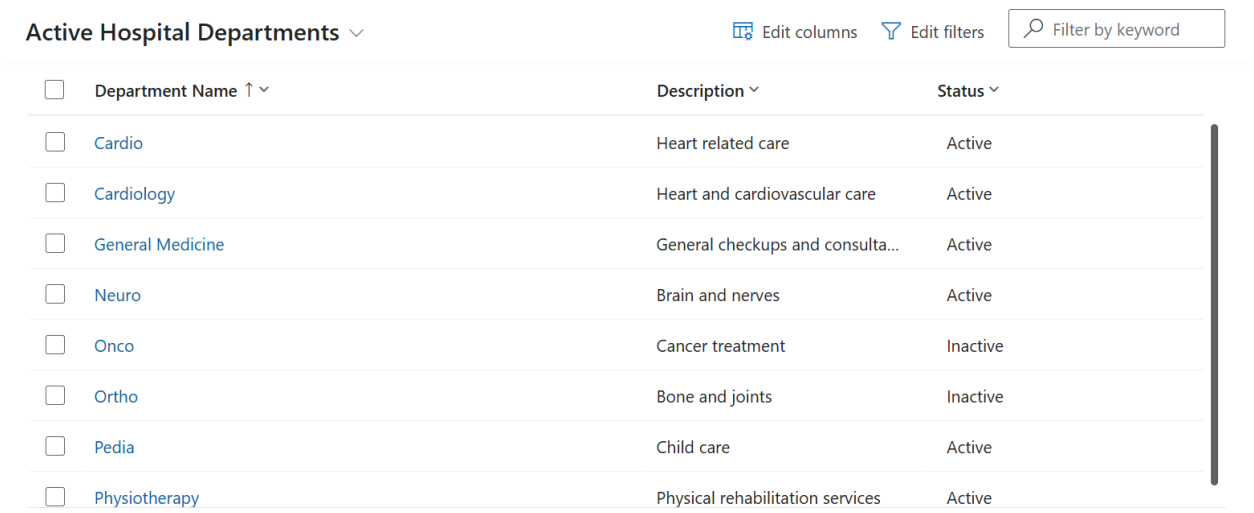


Fig 13. Hospital Department View

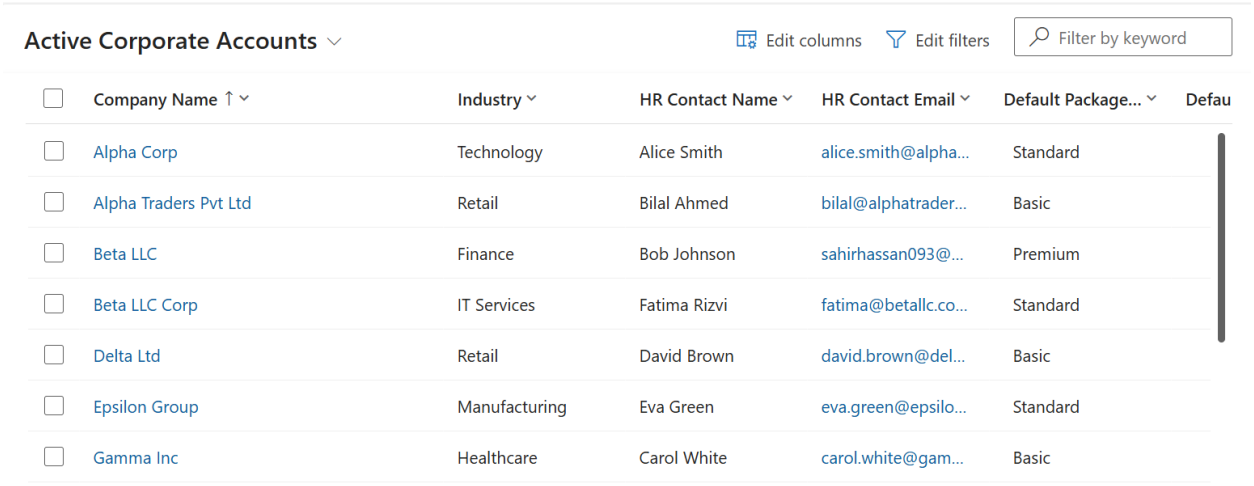


Fig 14. Corporate Accounts View

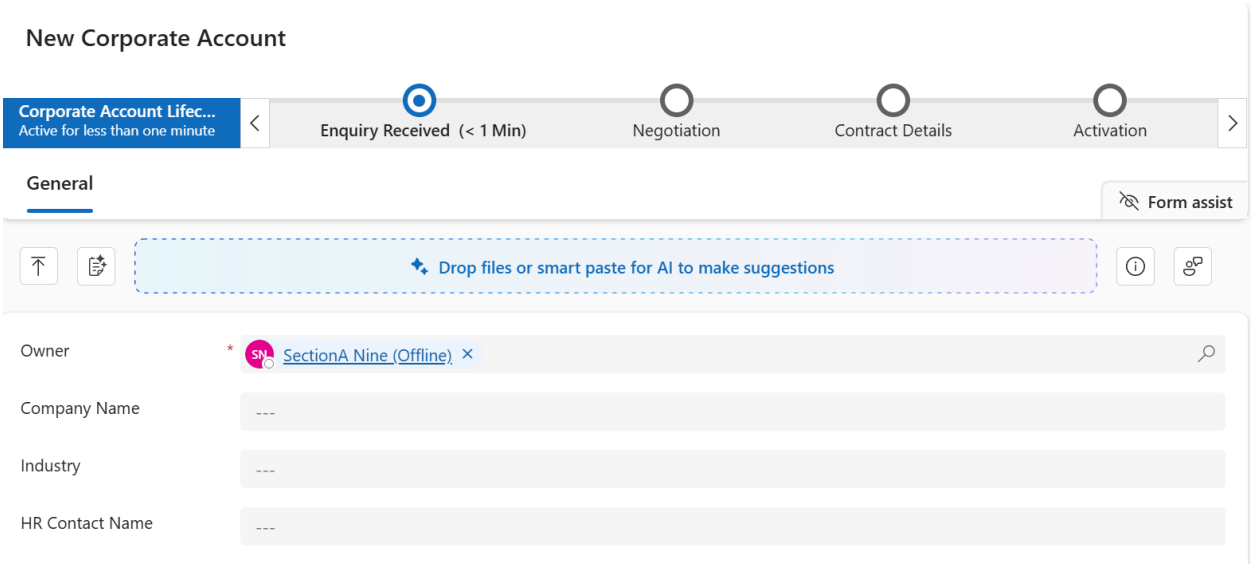
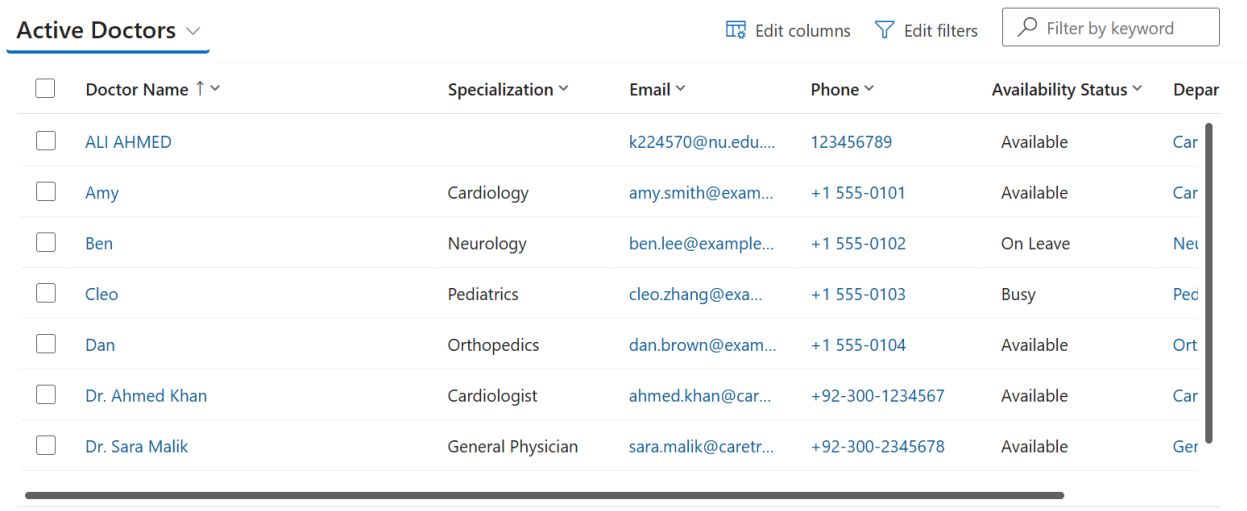
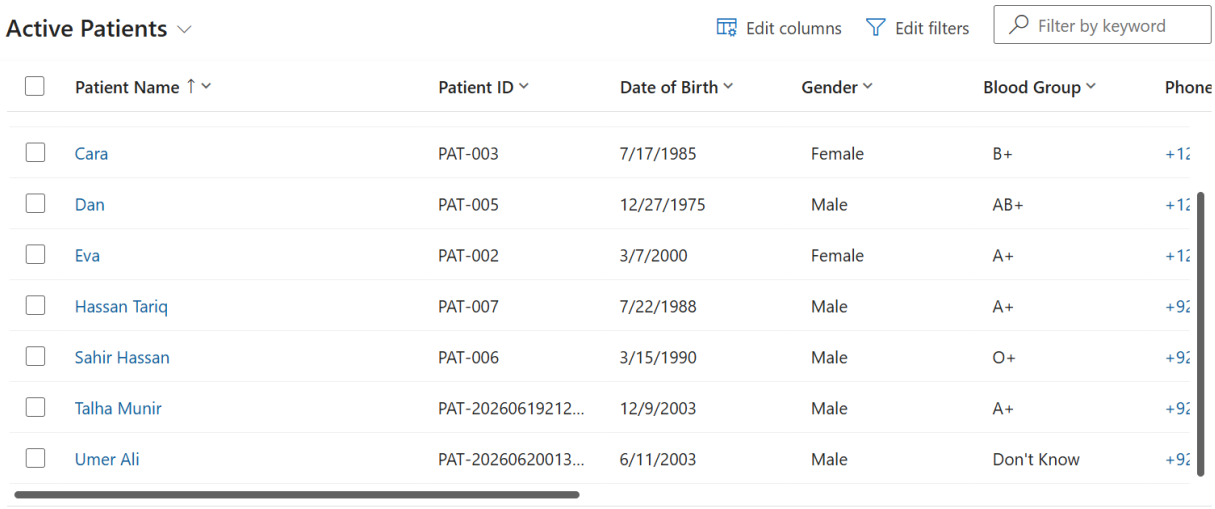


Fig 15. Corporate Accounts Form



Doctor Name	Specialization	Email	Phone	Availability Status	Depar
ALI AHMED		k224570@nu.edu....	123456789	Available	Car
Amy	Cardiology	amy.smith@exam...	+1 555-0101	Available	Car
Ben	Neurology	ben.lee@example...	+1 555-0102	On Leave	Neu
Cleo	Pediatrics	cleo.zhang@exa...	+1 555-0103	Busy	Ped
Dan	Orthopedics	dan.brown@exam...	+1 555-0104	Available	Ort
Dr. Ahmed Khan	Cardiologist	ahmed.khan@car...	+92-300-1234567	Available	Car
Dr. Sara Malik	General Physician	sara.malik@caretr...	+92-300-2345678	Available	Ger

Fig 16. Doctor View



Patient Name	Patient ID	Date of Birth	Gender	Blood Group	Phone
Cara	PAT-003	7/17/1985	Female	B+	+12
Dan	PAT-005	12/27/1975	Male	AB+	+12
Eva	PAT-002	3/7/2000	Female	A+	+12
Hassan Tariq	PAT-007	7/22/1988	Male	A+	+92
Sahir Hassan	PAT-006	3/15/1990	Male	O+	+92
Talha Munir	PAT-20260619212...	12/9/2003	Male	A+	+92
Umer Ali	PAT-20260620013...	6/11/2003	Male	Don't Know	+92

Fig 17. Patients View

7. Business Process Flow — Corporate Account Lifecycle

7.1 Purpose

Beyond simple data storage, CareTrack models the actual business workflow that a sales/admin team follows when onboarding a corporate client — from initial enquiry through negotiation, contracting, activation, and ongoing renewal tracking. This is implemented as a native Dataverse Business Process Flow (BPF) attached to the Corporate Account table, appearing as

a horizontal stage stepper at the top of the record form and surfacing only the fields relevant to the current stage.

7.2 Stage Definitions

#	Stage	Category	Key Fields Surfaced
1	Enquiry Received	Qualify	Company Name, Industry, Number of Enrolled Employees, HR Contact Name/Email/Phone, Default Package Type
2	Negotiation	Develop	Default Package Type, Default Max Visits Per Month, Number of Enrolled Employees, Payment Frequency
3	Contract Details	Propose	Total Contract Value, Contract Start/End Date, Payment Status, Amount Paid So Far
4	Activation	Close	Is Active, Corporate Account confirmation
5	Active / Renewal	Close	Contract End Date, Payment Status, Amount Paid So Far, Number of Enrolled Employees

7.3 How the Business Process Flow Behaves on a Record

A Business Process Flow functions as a guided checklist layered on top of a normal Dataverse form, rather than a separate process engine. Concretely, on a Corporate Account record it behaves as follows:

- When a new Corporate Account is created — for example, via the Canvas App's corporate enquiry form — Dataverse automatically starts that record at Stage 1, Enquiry Received.
- The stepper bar at the top of the form shows all five stages with the current stage highlighted, so any staff member can see at a glance where a given company sits in the pipeline simply by opening the record.
- Below the stepper, only the current stage's fields are shown in a compact card — at Enquiry Received, staff see Company Name, Industry, and HR contact details, without being shown Contract Value or Payment Status before those are relevant.
- A Next Stage action appears once the visible fields are complete. Where a field is marked Required for that stage (for example, Total Contract Value at Contract Details), Dataverse blocks progression to the next stage until it is filled in — this is the actual enforcement mechanism behind the “required-field completeness” behavior described in Section 7.6.
- Moving to a later stage does not delete or hide earlier data — once a record advances from Negotiation to Contract Details, the Negotiation-stage fields remain saved on the record; they are simply no longer the default active card. Staff can click back onto any earlier stage in the stepper to revisit or edit those fields at any time.
- The stage categories (Qualify, Develop, Propose, Close) are a visual layer only — each renders as a small icon in the stepper so it reads like a standard CRM sales pipeline, but they do not themselves alter validation or behavior.

7.4 Process Flow Diagram

Activity Diagram:

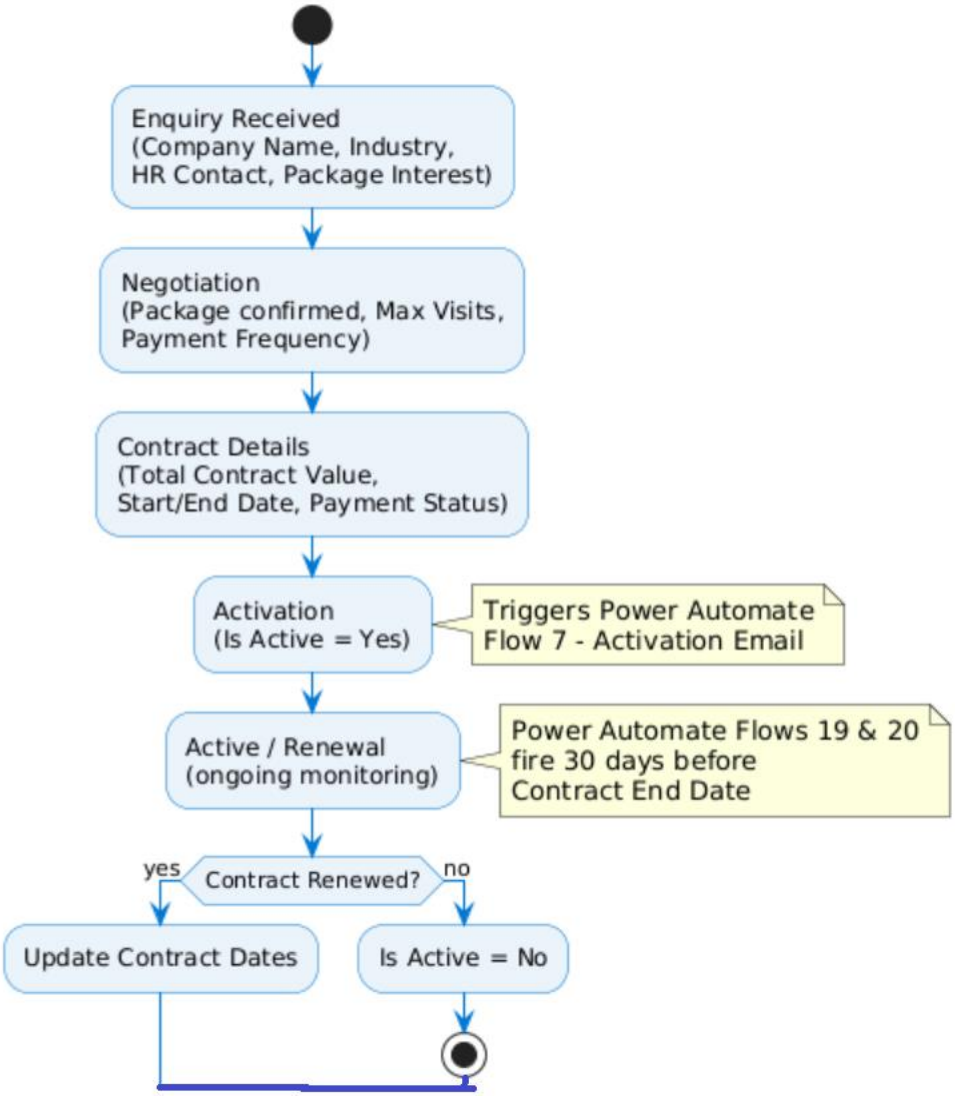


Fig 18. Activity Diagram for Business Process Flow for Corporate Account Lifecycle



Fig 19. Corporate Account Lifecycle Visible in Form

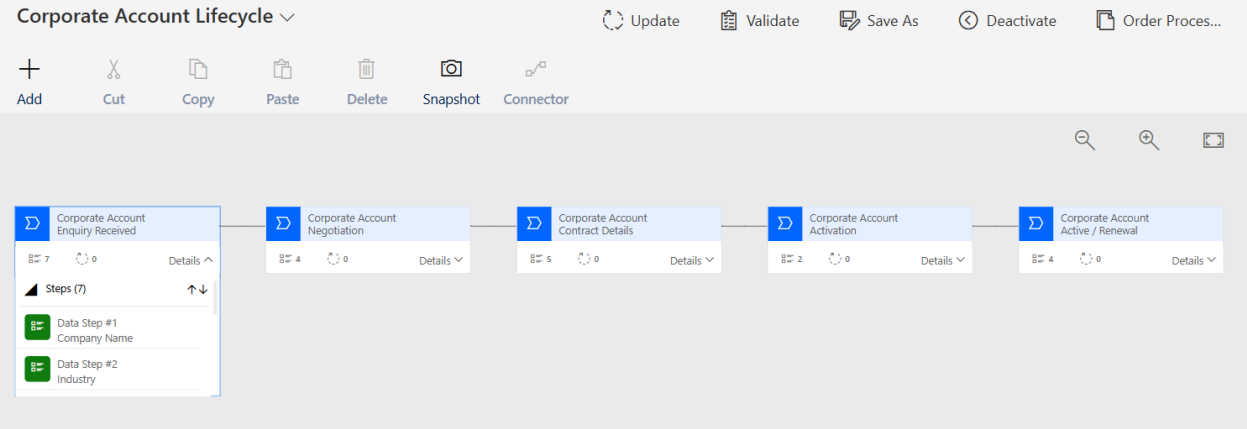


Fig 20. Corporate Account lifecycle with its steps to be fullfilled

7.5 Relationship to Power Automate — Parallel, Not Triggered

It is worth being precise about how the Business Process Flow relates to the Power Automate flows in Section 9: the BPF itself does not trigger any flow. The two operate side by side rather than through each other. Each relevant flow is watching a specific field directly, completely independent of which BPF stage the record happens to be sitting in at the time:

Field Change	What Actually Fires	Independent of BPF Stage?
Is Active changes to Yes (typically set while a record is at the Activation stage)	Flow 7 — Package Activation Notification, because Flow 7's trigger watches the Is Active column directly	Yes — fires on the field change alone, with or without the BPF present
Contract End Date approaching (checked daily)	Flows 19 & 20 — Contract Expiry Alerts (Admin and HR)	Yes — runs on a daily recurrence comparing dates, regardless of stage

In short, the BPF is a UI/UX layer that guides staff through the process and enforces field completeness at each step; the Power Automate flows are the actual automation engine reacting to underlying data changes. They achieve a coherent end-to-end experience by both being attached to the same record and the same fields — not because one calls the other.

7.6 Business Value of This Approach

Modeling the corporate account lifecycle as a Business Process Flow — rather than a flat record with a single Is Active toggle — demonstrates several things: required-field enforcement at each stage prevents incomplete records from progressing (e.g., a contract cannot reach Activation without a Total Contract Value); the stage categories (Qualify, Develop, Propose, Close) mirror standard CRM sales terminology, showing alignment with established business process patterns; and the visual stepper gives any staff member an immediate sense of where a given corporate relationship stands without needing to interpret multiple disconnected fields. We got inspiration from sales module opportunity pipeline.

8. Canvas App — CareTrack Patient Portal

8.1 Purpose and Design System

The CareTrack Patient Portal is a Tablet-format (1366 × 768) Canvas App used by patients, prospective family members, and corporate HR contacts to interact with the hospital without staff intervention. It covers patient registration, identity lookup (including multi-profile support for shared phone numbers), appointment booking against real-time doctor availability, appointment history and billing visibility, and a self-service corporate wellness enquiry funnel.

A consistent design system is applied across every screen to ensure a cohesive, professional appearance:

Element	Hex Value	Usage
Primary Blue	#0078D4	Headers, primary buttons, accents
Dark Blue	#005A9E	Hover states
Light Blue	#EBF3FB	Highlighted panels, callouts
Light Grey	#F5F5F5	Form card backgrounds
Border Grey	#D0D0D0	Input borders, dividers
Dark Text	#333333	Primary body text
Medium Grey	#666666	Secondary / helper text
Error Red	#FF0000	Validation error messages
Success Green	#00AA00	Success states, paid invoice status

Typography uses Segoe UI throughout, with a consistent hierarchy: 24pt bold for screen titles, 14pt italic for subtitles, 12–13pt for field labels and body content, matching the visual language of the Model-Driven App for a unified product feel across both applications.

8.2 Screen Inventory

Screen	Purpose
WelcomeScreen	Entry point — routes to booking, appointment history, registration, or corporate enquiry.
RegisterScreen	New patient registration with field validation and conditional insurance logic.
PatientLookupScreen	Phone-number identity lookup for returning patients.
PatientProfileSelectionScreen	Gallery of all patient profiles linked to a phone number (family members / multiple billing contexts).
ConfirmPatientDetailsScreen	Confirms the selected profile before proceeding to booking.
BookAppointmentScreen	Department → doctor → date → time slot booking flow, driven by real Doctor Schedule data.
ConfirmationScreen	Post-booking summary screen.

Screen	Purpose
MyAppointmentsScreen	Searchable, filterable gallery of the patient's appointment history.
AppointmentDetailScreen	Full appointment detail plus linked invoice/billing information.
CorporateEnquiryPackageScreen	Package overview / pitch page for prospective corporate clients.
CorporateEnquiryFormScreen	Corporate wellness enquiry submission form.
CorporateEnquiryConfirmationScreen	Enquiry submission acknowledgement.
OTPVerificationScreen	One-time-password identity verification layer.

8.3 WelcomeScreen

The landing screen presents three primary actions — Book an Appointment, My Appointments, Register as New Patient — alongside feature highlight panels (Multi-Department Care, Expert Doctors, Corporate Wellness Plans, Easy Booking, Digital Prescriptions, Secure & Private) and a subtle entry point into the corporate enquiry funnel.

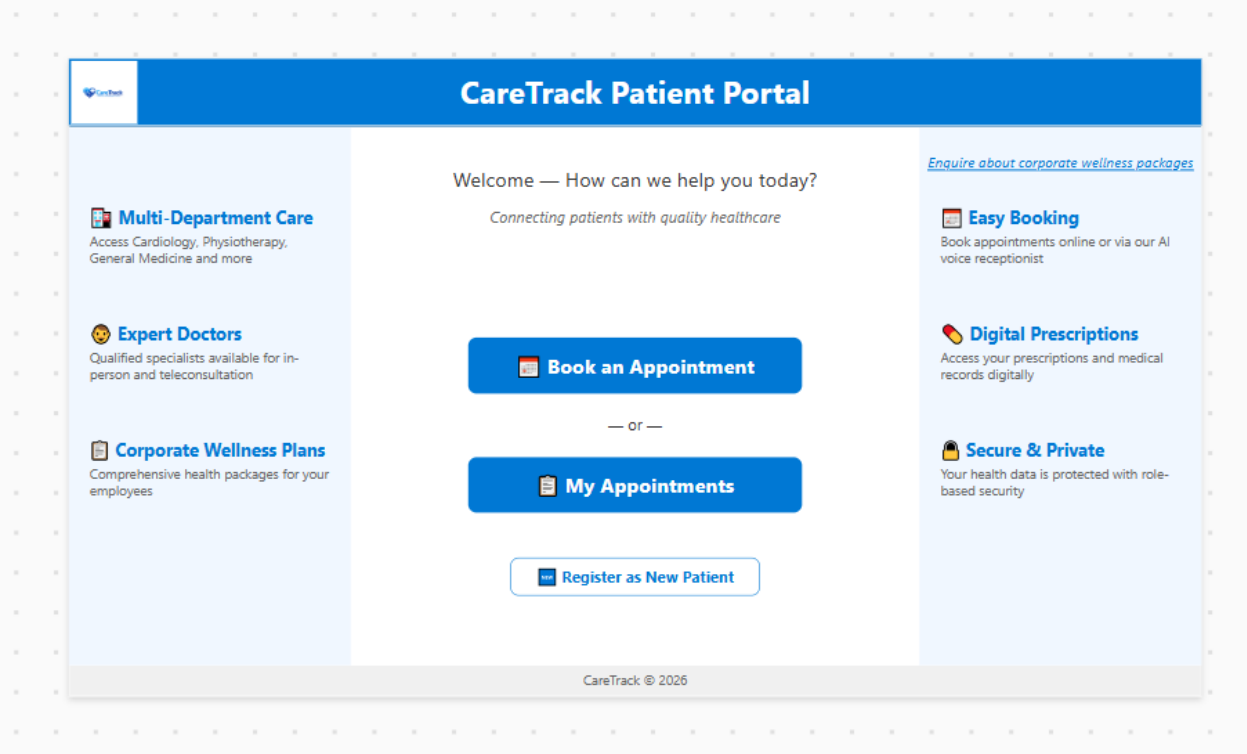


Fig 21. Home

8.4 RegisterScreen

Captures Personal Information (name, date of birth, gender, blood group) and Health Information (phone, email, patient type, medical history, insurance) in a two-column layout. Submission is guarded by layered validation logic before any record is written to Dataverse:

1. Required-field check: Full Name, Phone Number, Gender, and Patient Type must be present, or submission is blocked with an explanatory error.
2. Insurance consistency check: if an Insurance Provider is entered, a matching Policy Number is required, and vice versa — preventing incomplete insurance records.
3. On success, a new Patient record is created via Patch, a unique Patient ID is generated, the newly created record is captured directly from the Patch result (avoiding any race condition with concurrent registrations), and the patient proceeds to identity verification before booking.

Blood Group is optional and defaults to "Don't Know" at the Dataverse column level if left unselected — distinguishing a deliberate non-answer from an overlooked field.

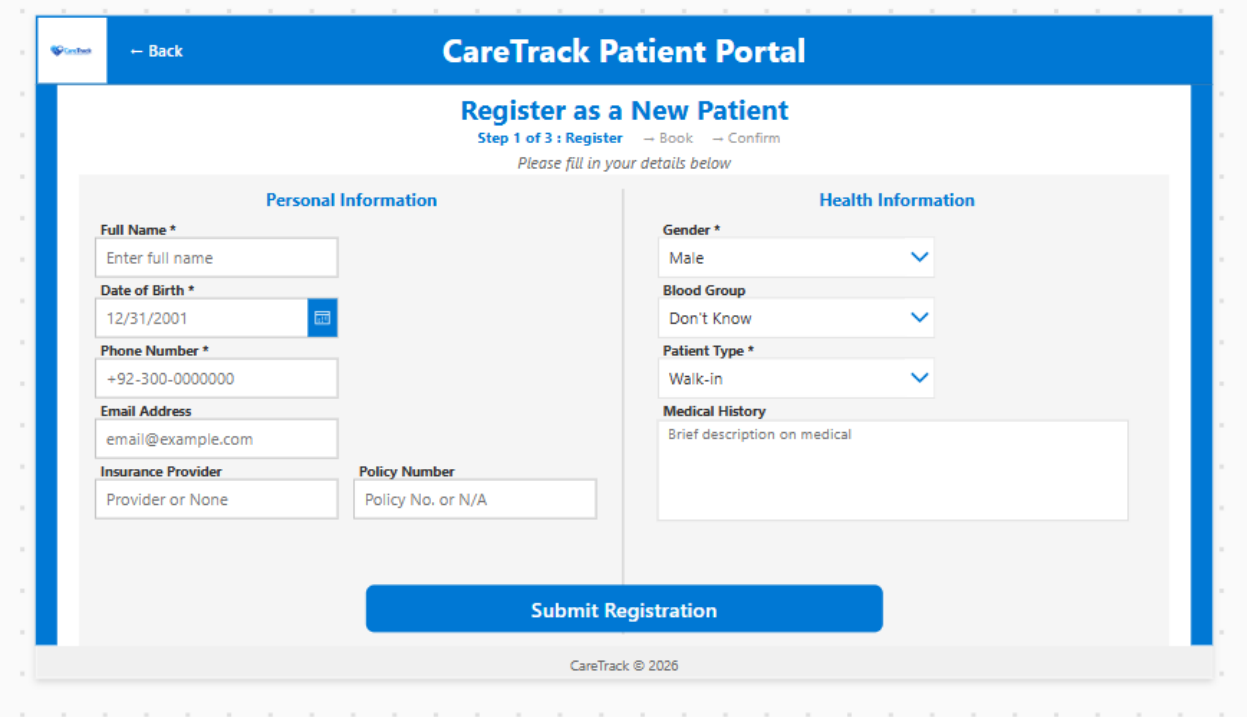


Fig 22. Create Patient

8.5 PatientLookupScreen

Returning patients identify themselves by phone number rather than a traditional username/password login, reflecting how walk-in clinics commonly operate. A successful lookup proceeds to identity verification (Section 8.13) and then routes to either My Appointments or Book Appointment, depending on which action the patient originally selected on the Welcome screen.

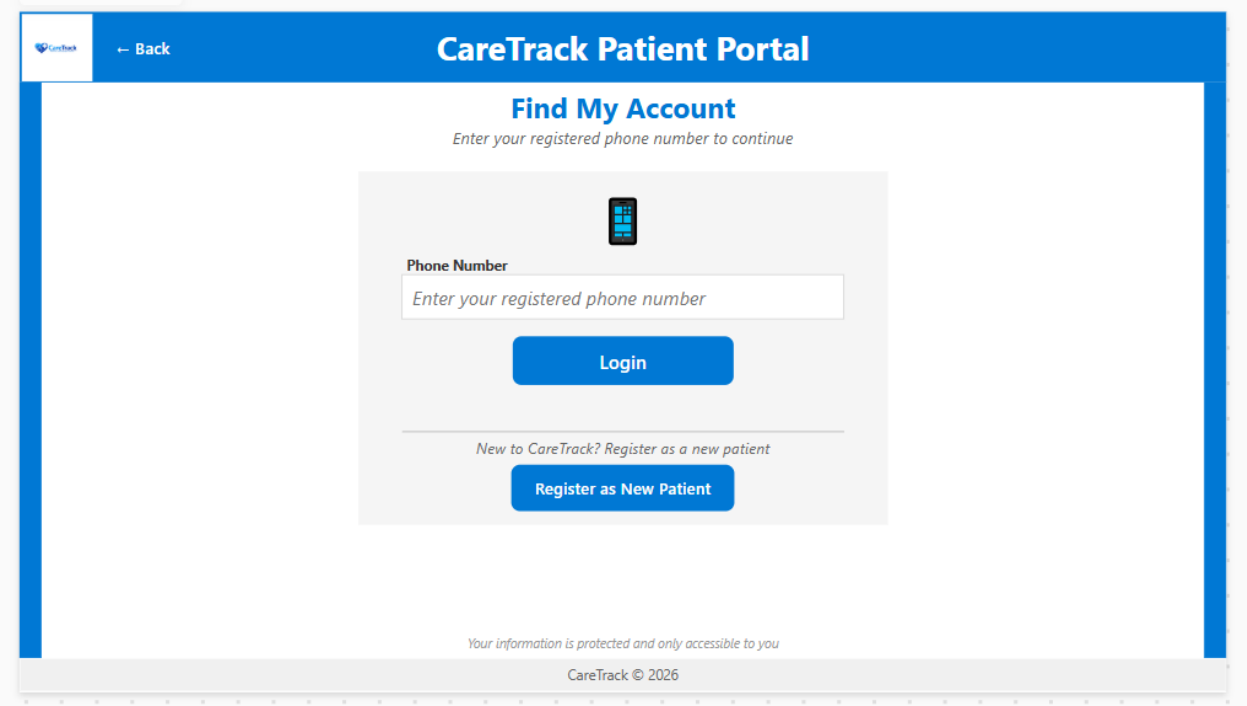


Fig 23. Login In

8.6 OTPVerificationScreen — Identity Verification Patient

Phone-number lookup alone is not a sufficient security boundary — anyone who knows a patient's phone number could otherwise view their appointment and medical history. To address this, a one-time-password (OTP) verification step sits between identity lookup and access to appointment data. After a successful phone lookup (or fresh registration), a four-digit code is generated and delivered through a channel the patient selects — email or SMS — via a corresponding Power Automate flow (Section 9.8). The patient must correctly enter this code before proceeding to their appointments or booking flow, and may request a new code if needed.

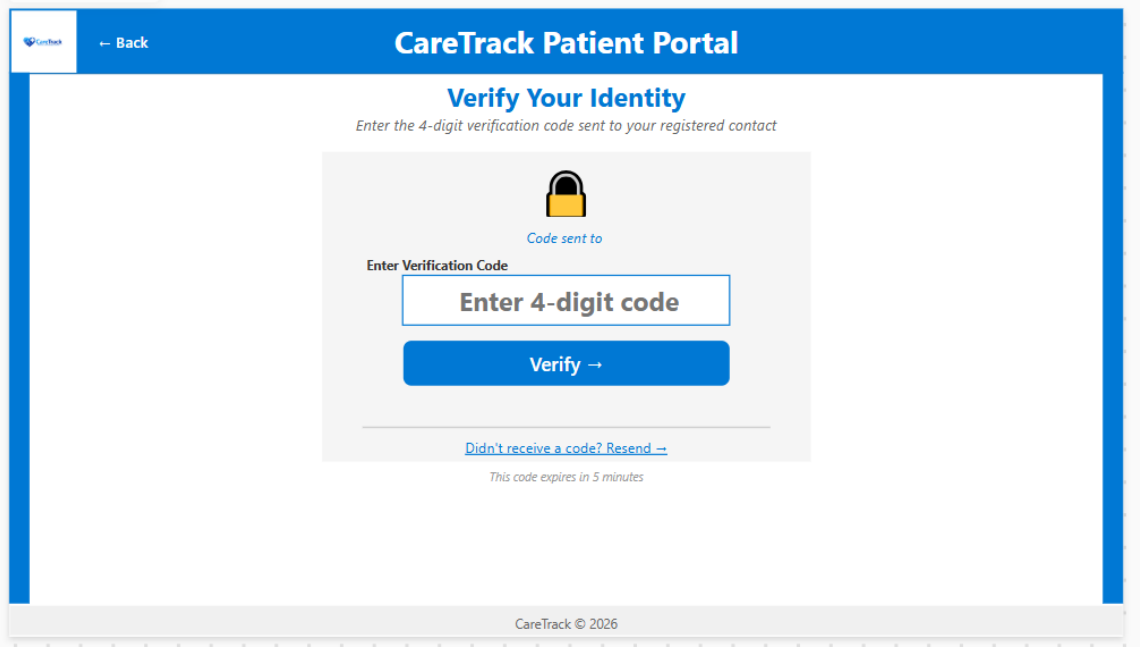


Fig 24. OTP via Email

8.7 ProfileSelectionScreen & ConfirmPatientDetailsScreen

A single phone number frequently maps to more than one patient — a parent registering a child, an elderly dependent without their own phone, or even the same individual holding separate Walk-in and Corporate records for different billing contexts (for example, using a company wellness package for routine visits but paying independently for a specialist outside that package). Rather than forcing a rigid one-phone-to-one-patient assumption, the portal surfaces every matching profile in a searchable gallery, displaying name, age, gender, and — for corporate records — the linked company name, before the patient confirms which profile to proceed with.

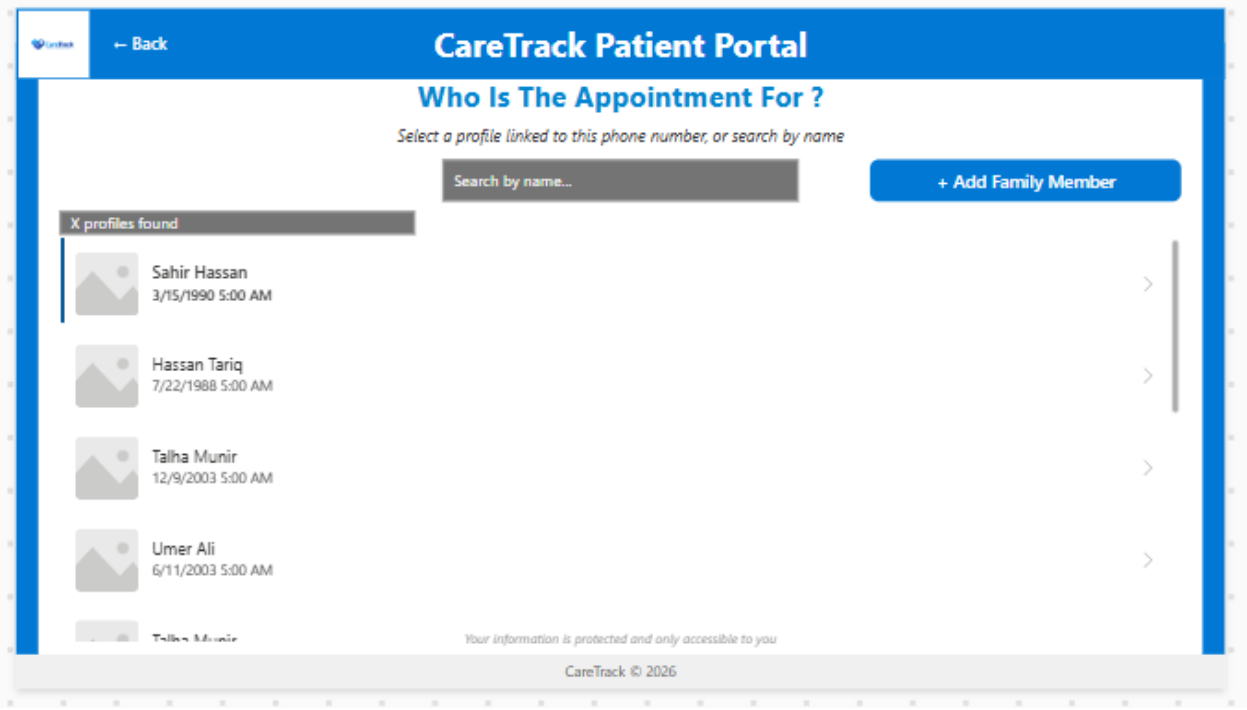


Fig 25. Multiple User Selection on one Phone Number

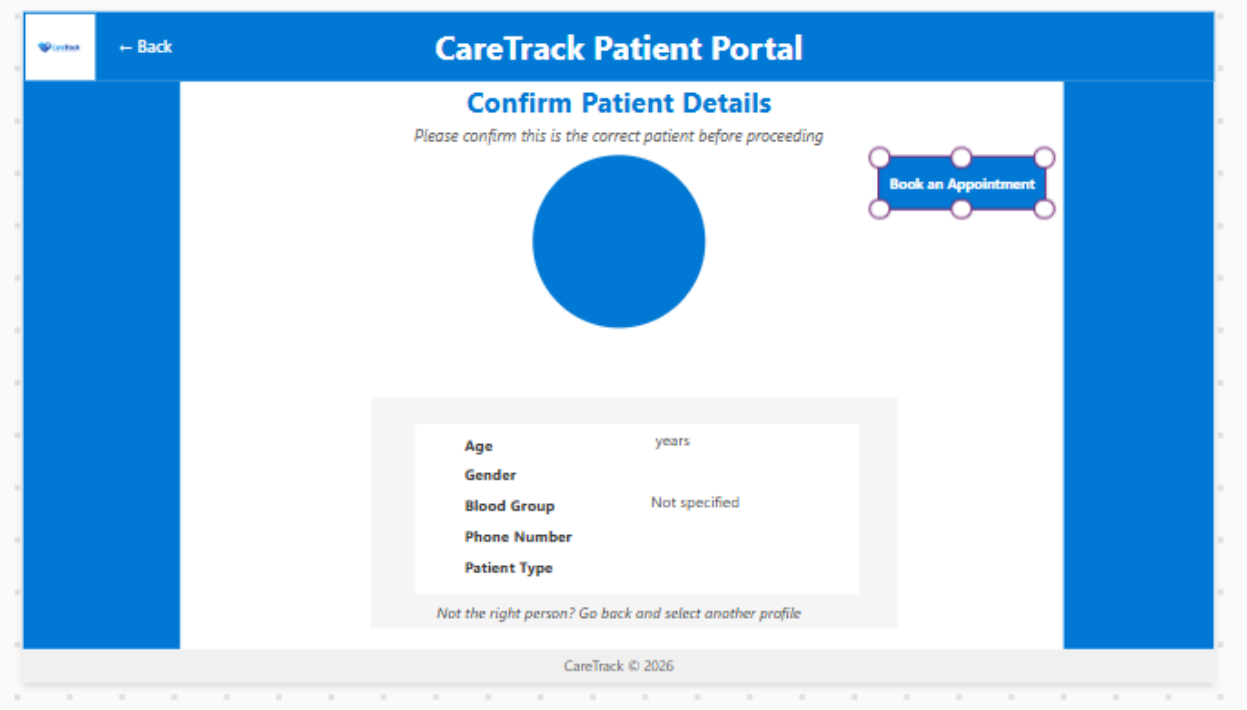


Fig 26. Select the User you want to perform the operation with

8.8 BookAppointmentScreen

The most operationally significant screen in the portal. Patients select a Department, which dynamically filters the Doctor dropdown to only that department's doctors (correctly handling the case of identically named doctors practicing in different departments, since filtering is keyed on the doctor's department relationship rather than name alone). Selecting a doctor and date triggers a multi-stage availability calculation:

1. All existing appointments for the selected doctor on the selected date are retrieved and converted into a list of already-booked time slots.
2. The doctor's working-hours schedule for that day of the week is retrieved from the Doctor Schedule table (which may include multiple records per doctor per department, supporting doctors who work different hours on different days).
3. A full list of 15-minute time slots is generated and filtered against the doctor's actual start time, end time, and break period — ensuring patients never see slots outside working hours or during a doctor's break.
4. Already-booked slots are removed from the remaining candidate list, leaving only genuinely available appointment times in the Preferred Time dropdown.

On confirmation, a new Appointment record is created with a generated appointment name combining patient name and department, the doctor's availability is re-validated for the selected day, and the patient proceeds to a confirmation summary.

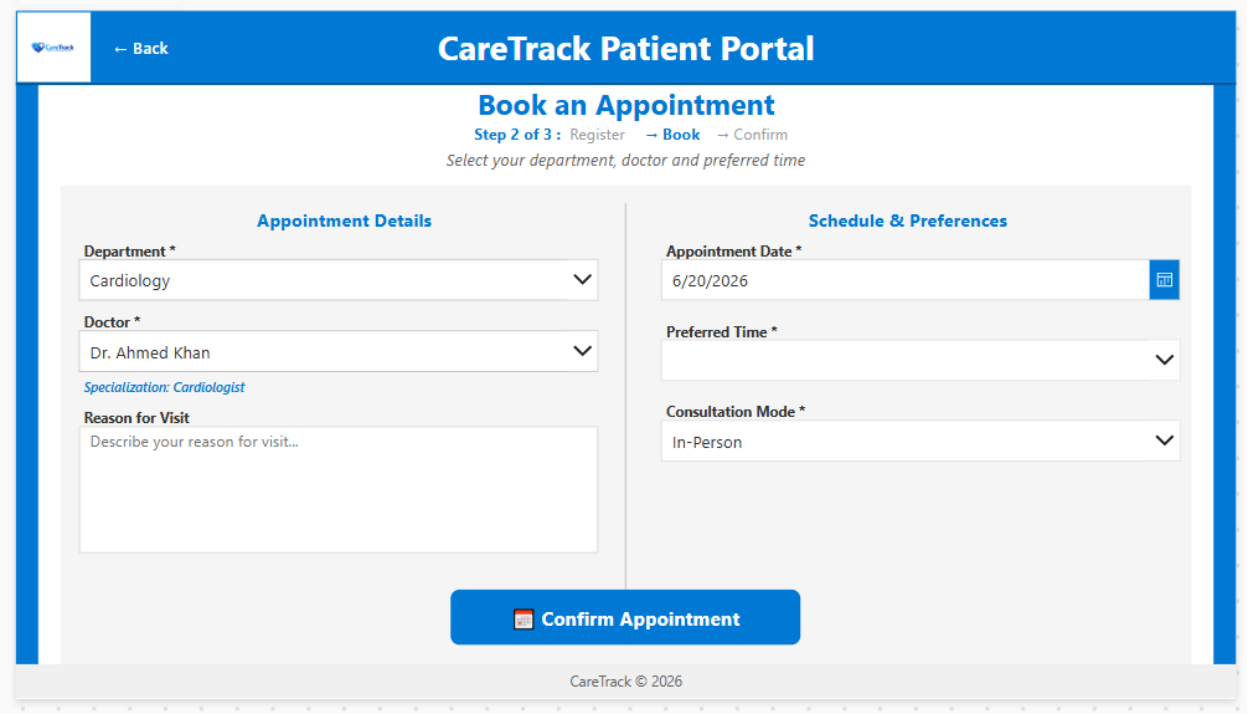


Fig 27. Booking Appointment

8.9 ConfirmationScreen

Displays a summary of the just-completed booking (patient, department, doctor, date, time, consultation mode) and offers two next actions: book another appointment, or navigate directly to the appointment history screen.

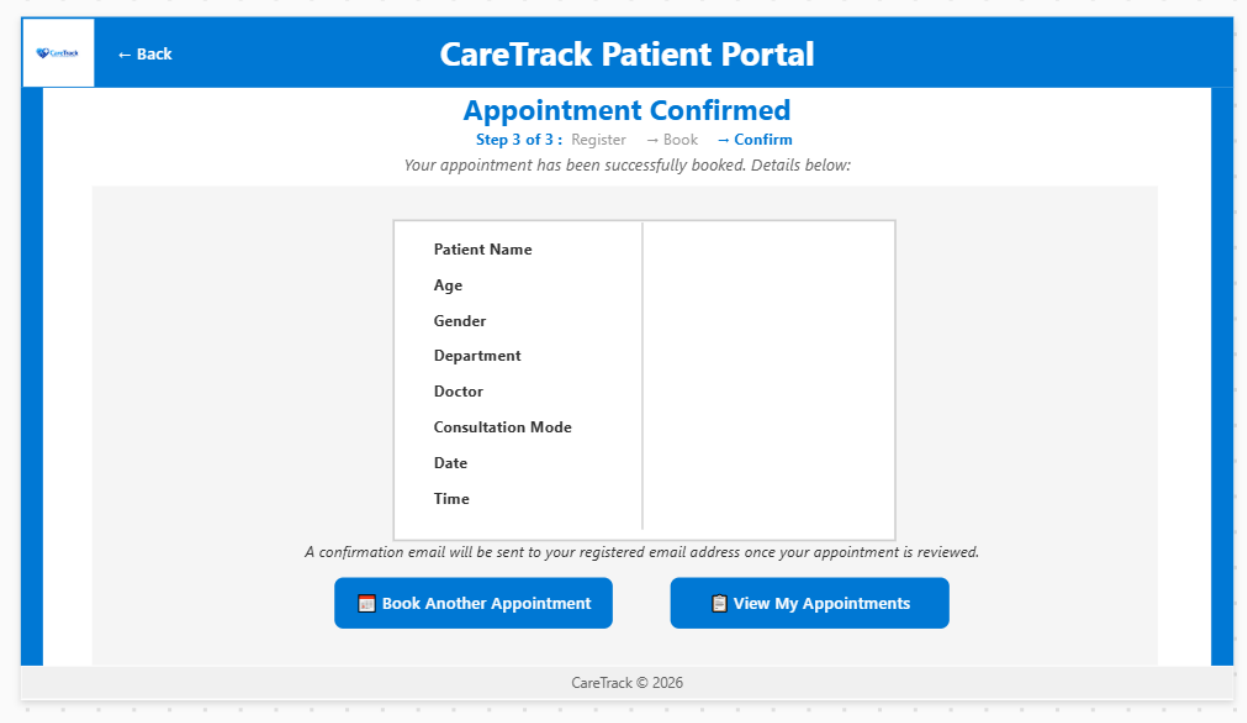


Fig 28. Confirmation & Details

8.10 MyAppointmentsScreen

A searchable, filterable gallery of the current patient's full appointment history, supporting free-text search (by doctor, department, or date) combined with a status filter (All, Scheduled, Completed, Cancelled, No Show). Each card displays the doctor, department, date/time, consultation mode, and a status badge color-coded for quick scanning. Selecting a card navigates to the full Appointment Detail view.

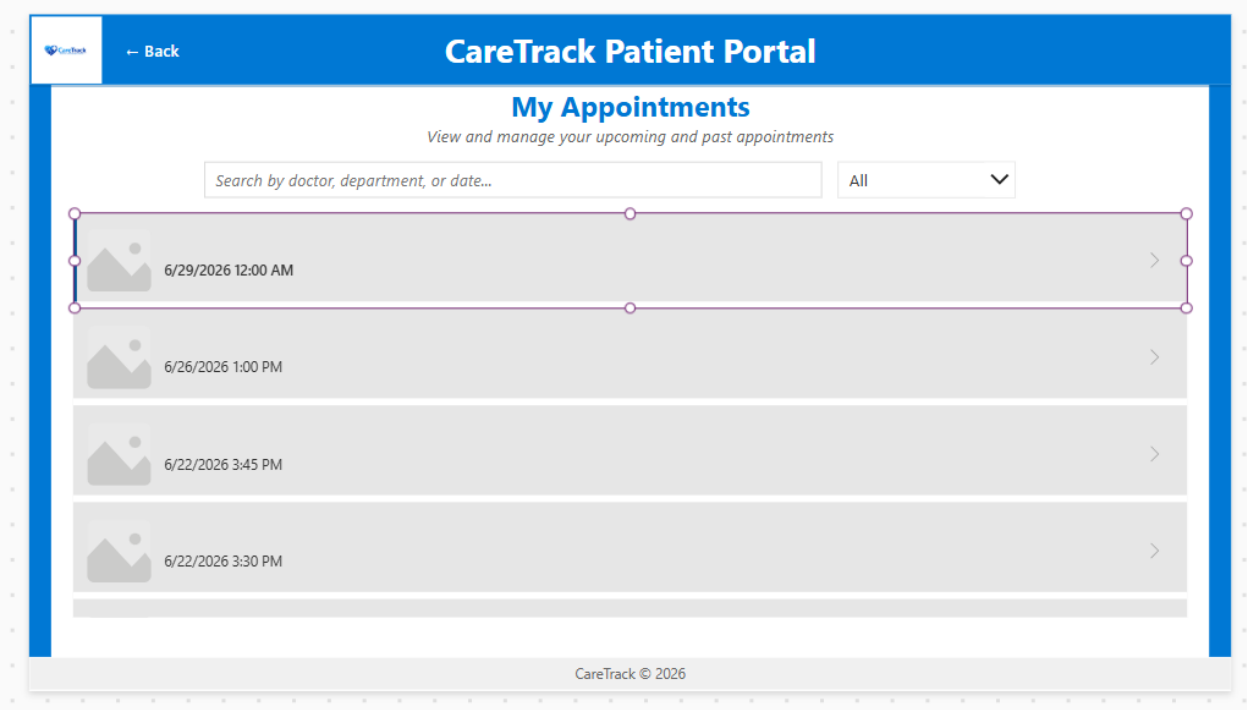


Fig 29. Older Appointments

8.11 AppointmentDetailScreen

Presents two side-by-side information panels for a selected appointment: Appointment Information (patient, age, gender, department, doctor, consultation mode, date, time, status) and Billing Information (invoice date, total amount, covered amount, patient payable amount, billed-to type, invoice status), the latter pulled from the linked Invoice record where one exists, or displaying “Pending” if the appointment has not yet been marked complete. A cancellation action is restricted to appointments still in Scheduled status.

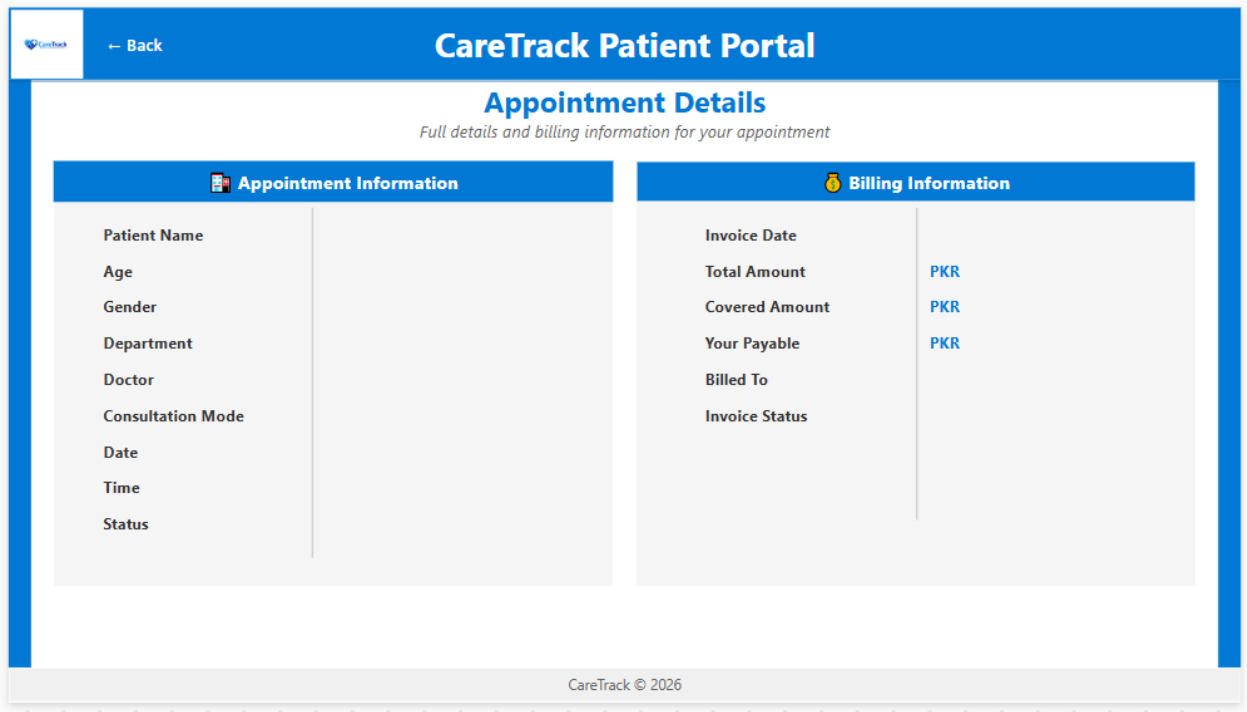


Fig 30. Appointment Details & Payment

8.12 Corporate Enquiry Funnel

A three-screen funnel allows prospective corporate clients (HR contacts) to self-serve an initial enquiry without requiring a phone call: a package overview/pitch screen (CorporateEnquiryPackageScreen) showing tiered package benefits at a glance, a submission form (CorporateEnquiryFormScreen) capturing company details, HR contact information, and package interest, and a confirmation screen acknowledging receipt and setting expectations for follow-up. Submitted enquiries create a new Corporate Account record with Is Active set to false, entering the Business Process Flow described in Section 7 at the Enquiry Received stage.

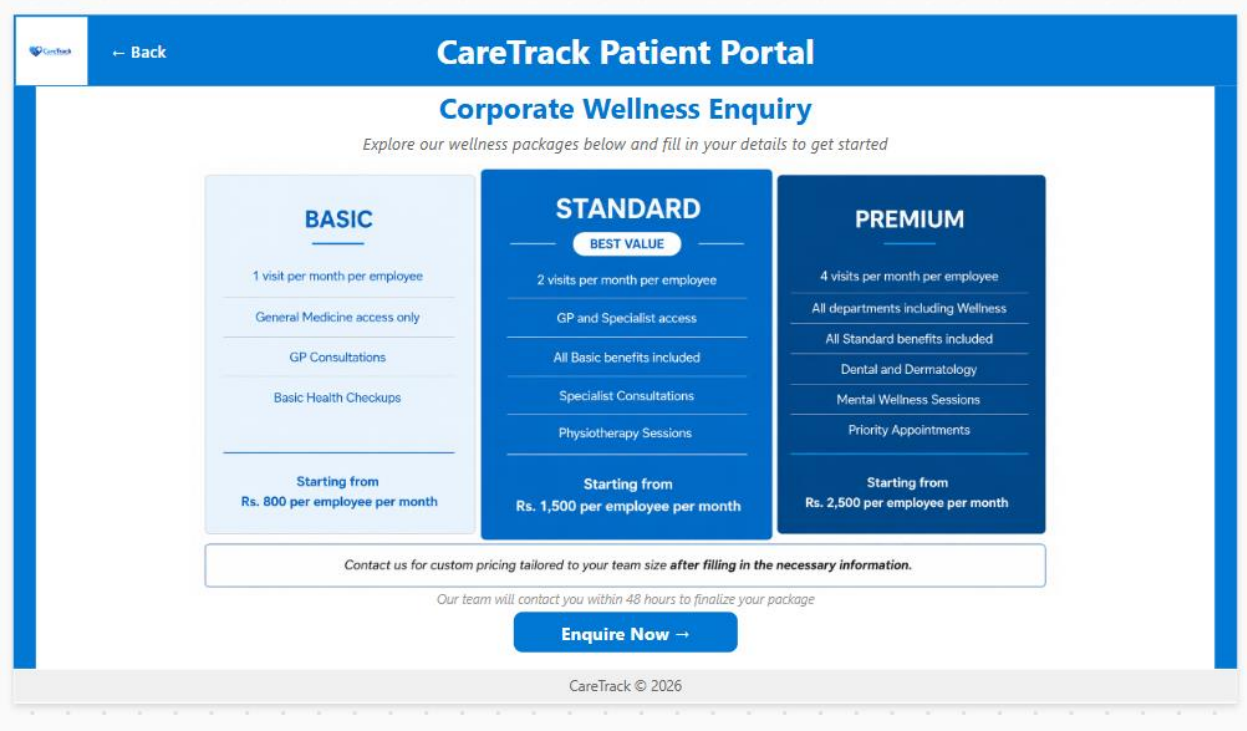


Fig 31. Package Types & Details

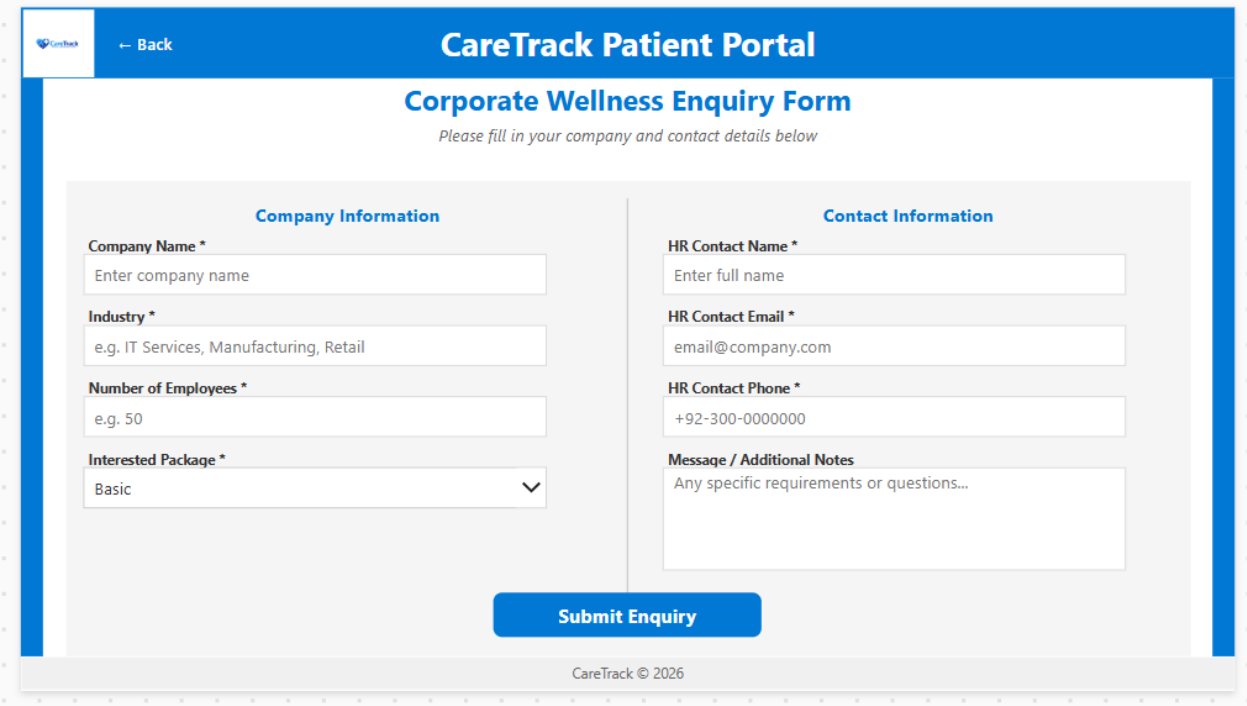


Fig 32. HR & Company Details through Enquiry Form

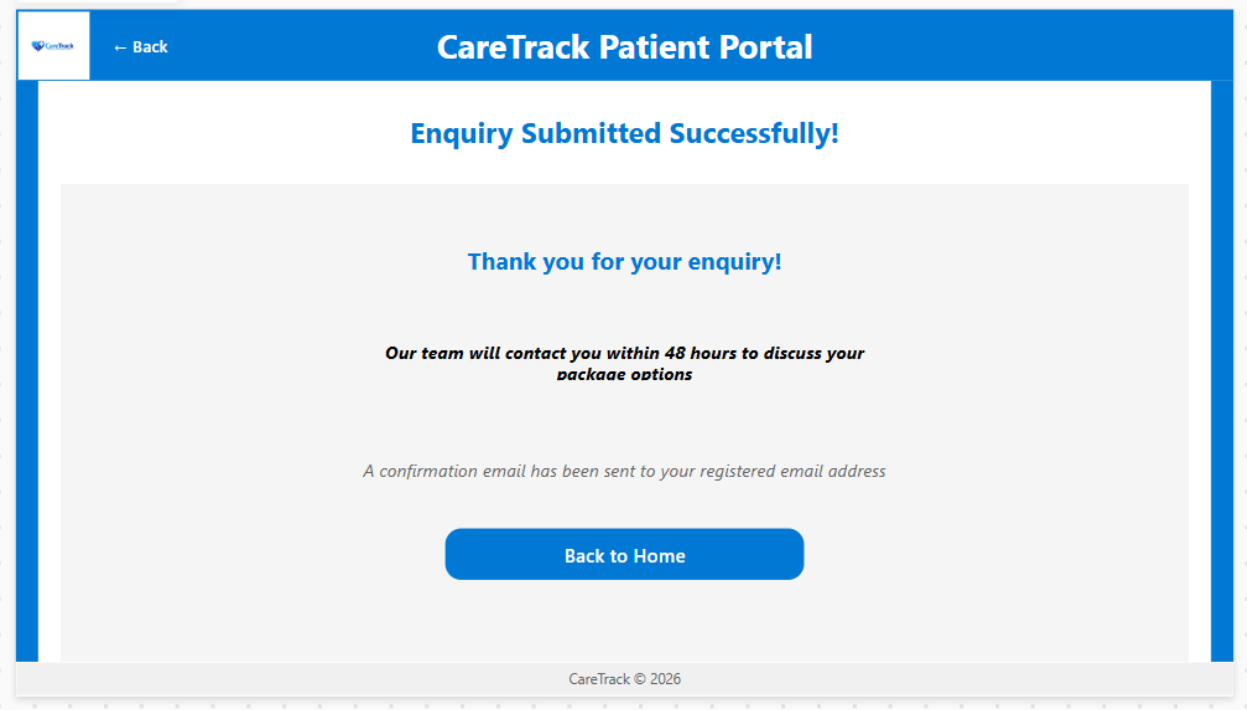


Fig 33. Success & Getting in Touch

8.13 Identity and "Login" Model — How Access Actually Works

CareTrack deliberately does not use a traditional username/password login. Walk-in healthcare access in practice does not work that way — patients identify themselves by phone number at a front desk, not by remembering a password for a hospital account they rarely use. The portal's identity model reproduces this pattern digitally while still protecting patient data:

1. The patient enters their registered phone number on PatientLookupScreen.
2. Dataverse is queried for every Patient record matching that phone number — not just one. If no match exists, the patient is prompted to register instead.
3. Rather than immediately granting access, a one-time password is generated and sent to the patient's chosen channel (email or SMS) through the OTP Power Automate flow (Section 9, Flow 8). This is the actual security boundary: knowing a phone number is not, by itself, sufficient to view someone's medical history or book on their behalf — the patient must also control the inbox or phone receiving the code.
4. Once the code is verified, if more than one Patient record shares that phone number, the patient is shown the profile selection gallery (Section 8.6) rather than being dropped directly into a single account — this is the mechanism that supports families and multi-context billing described below.

This design trades the familiarity of a conventional login for a flow that matches how patients already expect to interact with a clinic by phone, while still inserting a genuine verification step before any sensitive data is shown — rather than relying on phone number alone as an implicit password.

8.14 Appointment Slot Availability — How It Actually Works

The booking screen's time-slot logic is the most computationally involved part of the patient portal, and is worth explaining in full rather than leaving as a black box, since it directly determines whether a patient can ever double-book a doctor or be offered a time outside working hours.

1. Department → Doctor filtering: selecting a department filters the doctor list to only that department's doctors, matched on the doctor's department relationship rather than name — correctly handling the case where two different doctors happen to share the same name in different departments.
2. Schedule retrieval: once a doctor and date are chosen, every Doctor Schedule record belonging to that doctor for the matching day of the week is retrieved. A doctor may have more than one schedule record for the same day across different periods, or different hours on different days — the logic accounts for all matching records rather than assuming exactly one.
3. Candidate slot generation: a fixed list of time slots in 15-minute increments is generated across the full clinic-operating range, then filtered down to only those falling within the doctor's Start Time and End Time for that schedule record, with any slot inside the doctor's Break Start–Break End window explicitly excluded.
4. Conflict removal: every existing Appointment for that doctor on that date is retrieved and converted into a list of already-occupied times; these are subtracted from the candidate slot list, so a slot already booked by another patient never appears as selectable.
5. Past-time exclusion: if the selected date is today, slots earlier than the current time are also excluded, preventing a patient from booking an appointment time that has already passed.
6. Final validation on confirm: even after a slot is selected, the Confirm Appointment action re-checks that the doctor has a schedule record for the selected day before writing the Appointment record — a defensive second check in case the doctor's schedule changed between when the slot list was generated and when the patient confirmed.

Doctor Schedule Start Time, End Time, Break Start, and Break End are stored as plain text in 24-hour format (e.g., “10:00”) rather than a native time type, since Dataverse does not provide a time-only field independent of a date component. All comparisons described above are therefore performed by parsing these text values into total minutes since midnight at the application layer — a deliberate choice to avoid unpredictable timezone conversion behavior that a Date/Time-typed field would introduce purely for a recurring weekly schedule that has no calendar date of its own.

8.15 Use Case Walkthroughs

The following walkthroughs illustrate scenarios the identity and booking model above was specifically designed to support, beyond the simplest “one patient books for themselves” case.

Use Case 1 — Booking on Behalf of a Family Member

A parent has previously registered themselves, their child, and an elderly parent under their own phone number, since the child and grandparent do not have independent phones registered with the hospital. When the parent opens the portal and chooses “Book an Appointment,” they

look themselves up by phone number, verify via OTP, and are shown a gallery containing all three linked profiles — displayed with name, age, and gender so they are easy to tell apart at a glance. The parent selects the child's profile, confirms the details, and proceeds into the booking flow, which then books the appointment under the child's Patient record, department, and chosen doctor — not the parent's own record.

Use Case 2 — Same Person, Two Different Patient Types

An employee at a corporate client is enrolled in their company's wellness package as a Corporate patient. Earlier in the year, before their company's contract was activated, they had separately registered as a Walk-in patient and paid for a visit themselves. Both Patient records exist under the same phone number and the same name. When this employee looks themselves up, the profile gallery shows both records side by side, each carrying a Patient Type badge (Walk-in vs. Corporate) and, for the Corporate record, the linked company name as a subtitle — so the employee can deliberately choose which billing context applies to today's visit, rather than the system silently assuming one.

Use Case 3 — Corporate Benefit Exhausted Mid-Month

The same Corporate employee has already used their monthly visit allowance under their company's package (tracked via the Employee table's Visits Used This Month counter against their Max Visits Per Month, inherited from their Corporate Account's default unless individually overridden). They need a further visit this month. Rather than the system blocking the booking outright, the appointment can still be made under their Corporate Patient record — the distinction is handled at the billing stage: when the appointment is completed and the Invoice is auto-created (Section 9, Flow 4), the covered amount reflects that their package allowance is exhausted, and the patient-payable amount is calculated accordingly rather than being fully covered. The Low Package Visits Warning flow (Flow 16) proactively notifies the employee as they approach this limit, so the shift from covered to payable is not a surprise at billing time.

Use Case 4 — Booking for a Friend Without a Prior Relationship

Because profile selection is driven purely by phone number rather than a fixed household or family construct, the same mechanism that supports parents booking for children also supports any group of people who choose to share a contact number for hospital purposes — friends, roommates, or colleagues who register under one number for convenience. There is no hard-coded relationship type (such as “parent” or “child”) required between profiles sharing a phone number; the system only requires that the phone number matches, and the registering individual chooses how to label the relationship informally during in-person registration if needed.

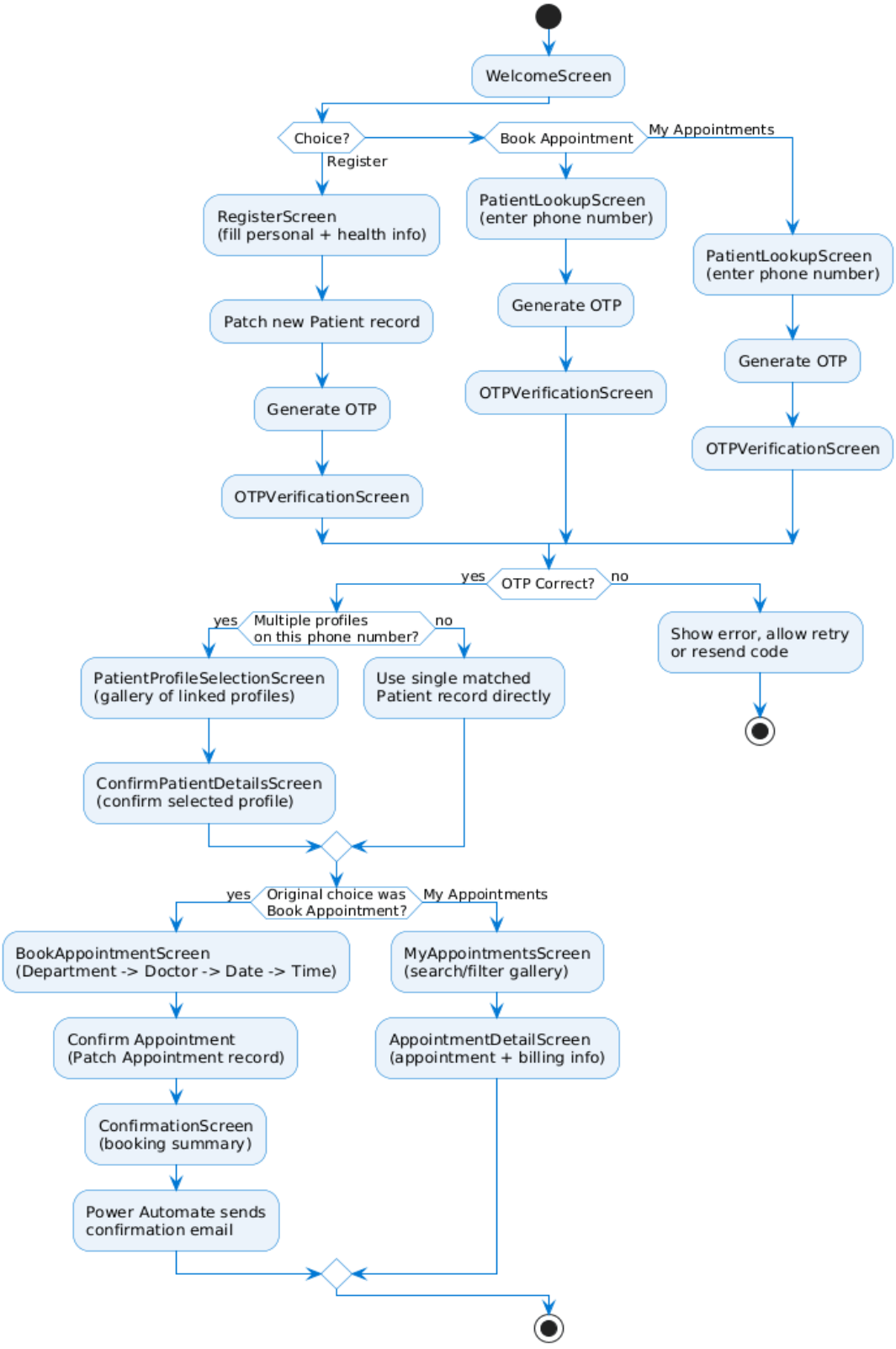


Fig. Patient Sequence Diagram

9. Power Automate — Process Automation

9.1 Automation Strategy

CareTrack's Power Automate layer is organized around the principle that every meaningful state change in the system — an appointment being booked, completed, or cancelled; an invoice being paid; a corporate account being activated or approaching contract expiry — should trigger an appropriate, automatic response, rather than relying on staff to remember to send a follow-up email or update a related record manually. Twenty-one flows were designed across the full patient and corporate account lifecycle, falling into four broad categories: patient communication, corporate account lifecycle automation, internal operational alerts, and record-creation automation that keeps the Medical Record and Invoice tables in sync with appointment status changes without manual data entry.

9.2 Flow Inventory

#	Flow	Trigger	Purpose
1	Appointment Confirmation Email	Row added (Appointment)	Emails patient booking confirmation with full appointment details.
2	Appointment Reminder	Recurrence (daily, 8 AM)	Reminds patients of appointments scheduled for the next day.
3	Medical Record Auto-Create	Row modified (Status = Completed)	Creates a blank Medical Record shell for the doctor to complete.
4	Invoice Auto-Create	Row modified (Status = Completed)	Creates a corresponding Invoice record with billing context.
5	Corporate Enquiry — HR Confirmation	Row added (Corporate Account)	Acknowledges receipt of a new corporate enquiry to HR.
6	Corporate Enquiry — Admin Alert	Row added (Corporate Account)	Notifies internal admin of a new enquiry requiring follow-up.
7	Package Activation Notification	Row modified (Is Active → Yes)	Confirms activation to HR and outlines onboarding next steps.
8	OTP Verification	PowerApps (manual trigger)	Delivers a one-time password via email or SMS for identity verification.
9	Invoice Paid Confirmation	Row modified (Status = Paid)	Sends a payment receipt email summarizing the invoice.
10	Visits Counter Increment	Row modified (Status = Completed, Corporate)	Increments the employee's monthly visit counter.
11	Monthly Visits Reset	Recurrence (1st of month)	Resets all employees' monthly visit counters to zero.
12	HR Negotiation Meeting Scheduler	Row added (Corporate Account)	Books a negotiation call with HR and sends a meeting link.
13	Teleconsultation Meeting Link	Recurrence (every 15 min)	Sends a video meeting link 30 minutes before online consultations.

#	Flow	Trigger	Purpose
14	Employee Bulk Import	File created (OneDrive)	Imports an HR-provided Excel employee list into the Employee table.
15	Patient No-Show Alert	Recurrence (hourly)	Flags appointments not marked complete 30+ minutes after their time.
16	Low Package Visits Warning	Row modified (Employee)	Warns an employee approaching their monthly visit limit.
17	Welcome Email on Registration	Row added (Patient)	Sends a welcome email confirming successful registration.
18	Appointment Cancellation Notification	Row modified (Status = Cancelled)	Notifies both doctor and patient of a cancellation.
19	Contract Expiry Alert — Admin	Recurrence (daily, 9 AM)	Alerts admin 30 days before a corporate contract expires.
20	Contract Expiry Alert — HR	Recurrence (daily, 9 AM)	Alerts the HR contact 30 days before their contract expires.
21	New Doctor Onboarding Notification	Row added (Doctor)	Reminds admin to configure a schedule for a newly added doctor.

9.3 Key Flow Walkthrough: Invoice Auto-Create

As a representative example of the automation pattern used throughout the system, the Invoice Auto-Create flow is detailed below.

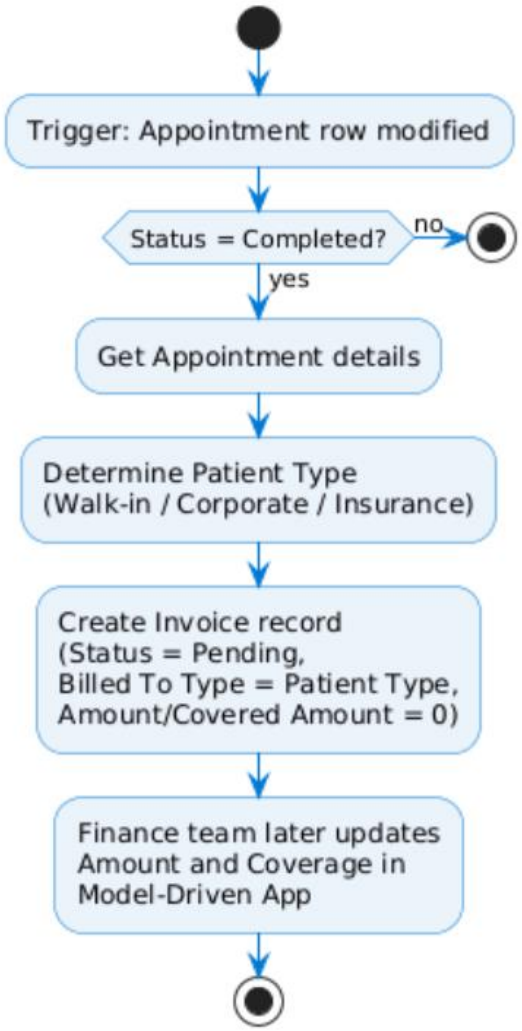


Fig 34. Invoice Power Automate Activity Diagram for CareTrack – Auto Create Invoice

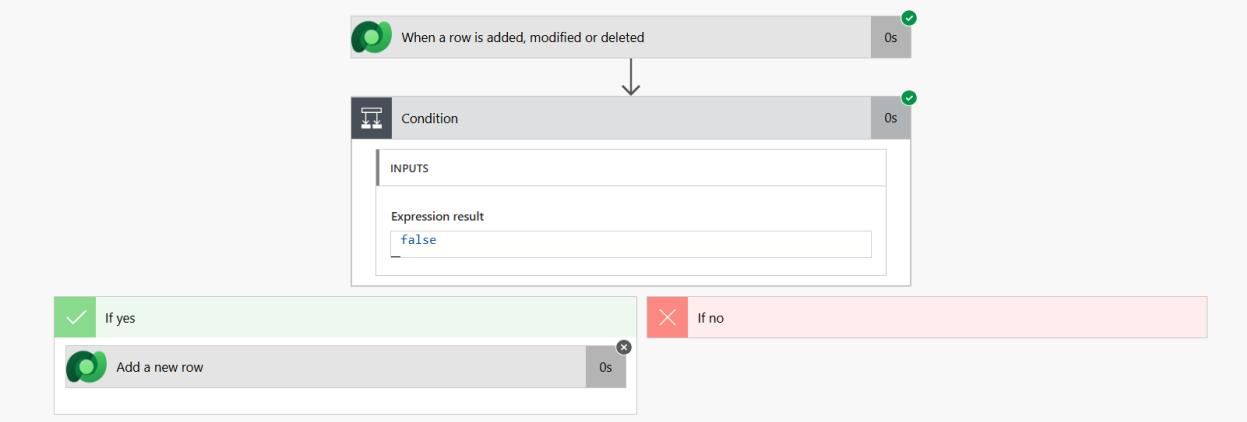


Fig 35. Power Automate Flow

9.4 Key Flow Walkthrough: OTP Verification

This flow is invoked directly from the Canvas App (rather than triggered by a Dataverse event) and demonstrates two-way integration between Power Automate and the patient portal.

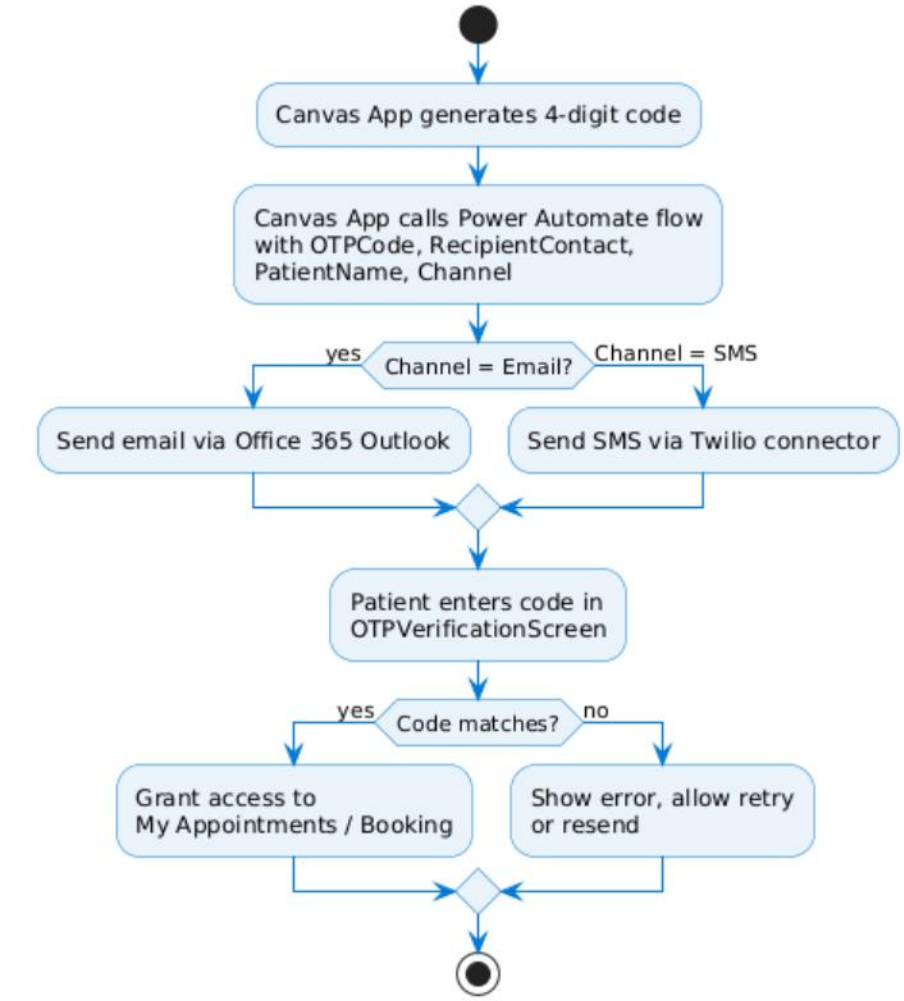


Fig 36. Initial Design for the Activity Diagram for CareTrack – Send OTP Email

Note this couldn't be implemented due to the paid feature.

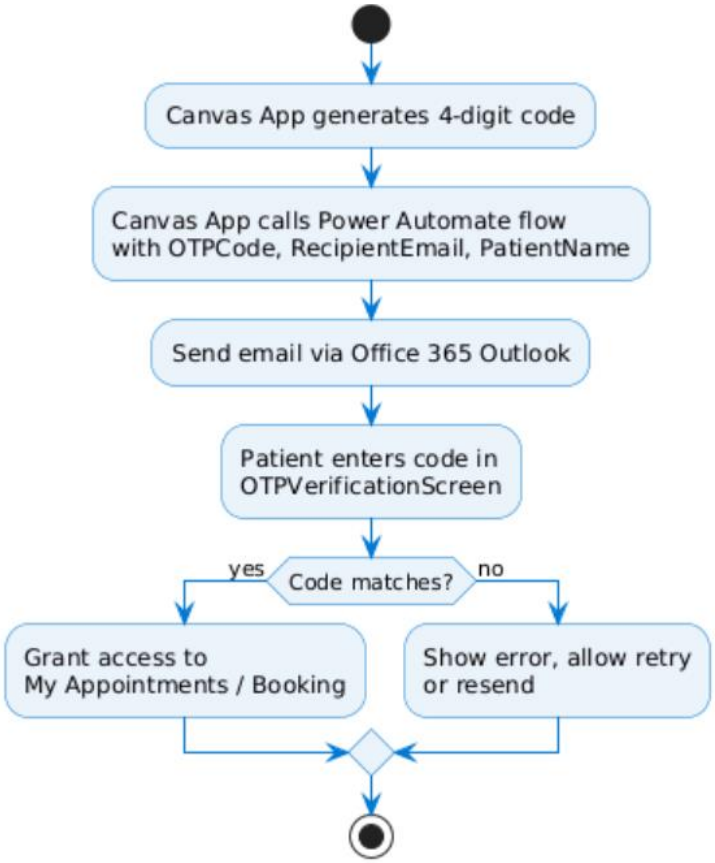


Fig 37. Activity Diagram for CareTrack – Send OTP Email

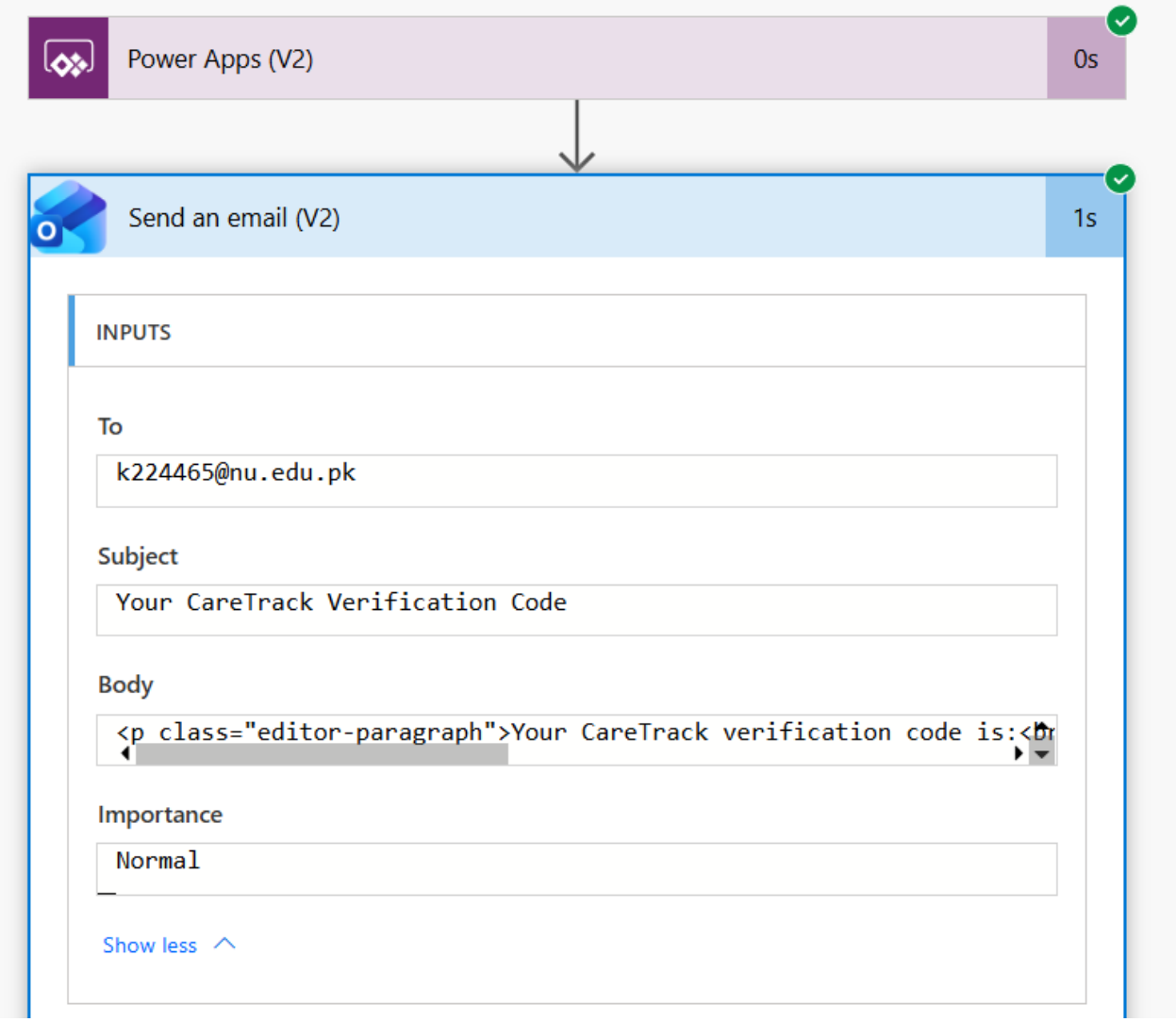


Fig 38. Power Automate Flow for CareTrack – Send OTP Email

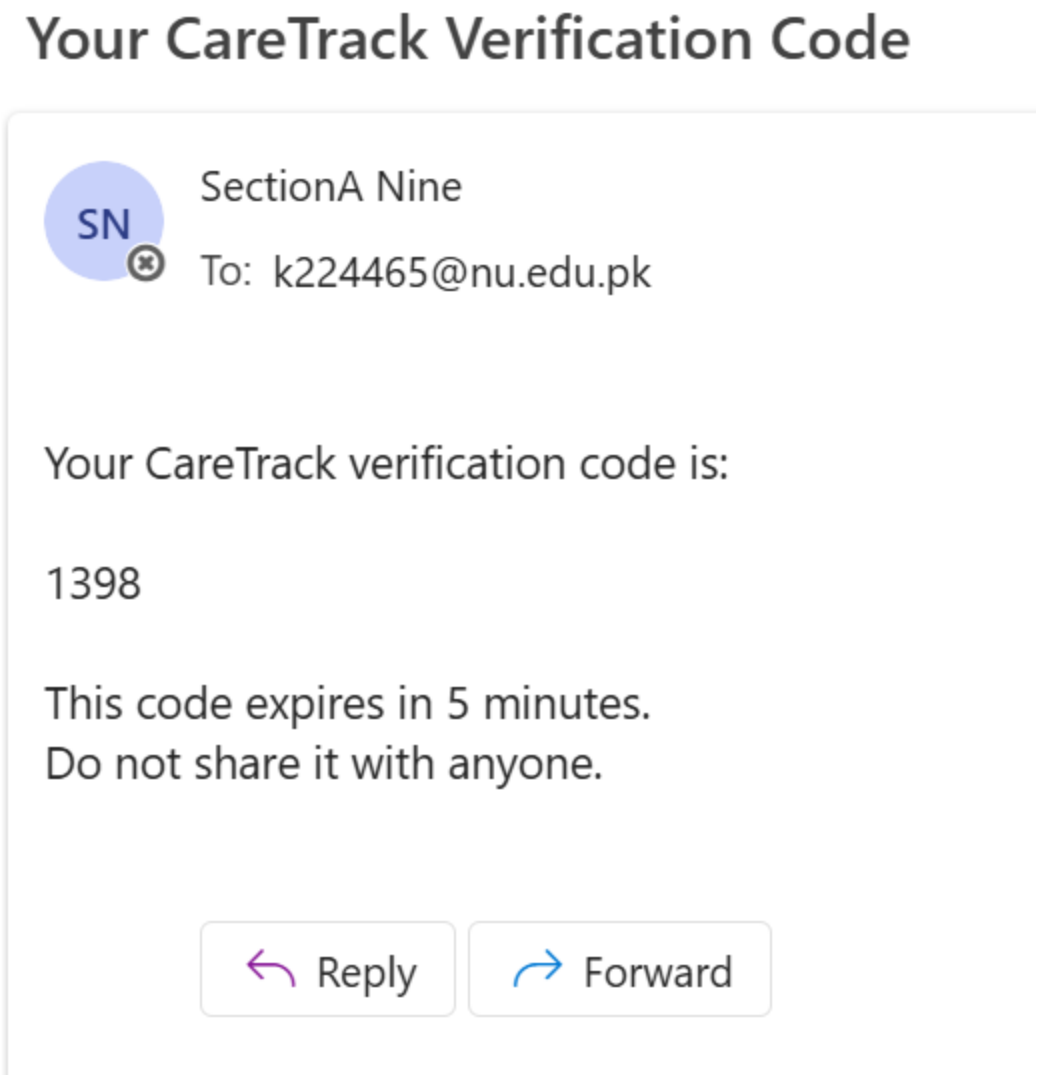


Fig 39. Email OTP Send to receiver for CareTrack – Send OTP Email

9.5 Connector Summary

Connector	Used For
Microsoft Dataverse	All record-based triggers (row added/modified) and CRUD actions across flows
Office 365 Outlook	All transactional and notification emails
Excel Online (Business) / OneDrive for Business	Employee bulk import from HR-provided spreadsheets
PowerApps (V2)	Manual trigger from the Canvas App for OTP delivery
Twilio (optional)	SMS delivery channel for OTP, where configured

10. Copilot Studio — CareTrack Booking Assistant

10.1 Purpose and Scope

Not every patient is comfortable navigating a multi-screen booking form. CareTrack includes a Copilot Studio conversational agent — the CareTrack Booking Assistant — that lets a patient describe a symptom, receive a department recommendation, and book a real appointment entirely through natural-language chat, reading and writing the same Dataverse tables that power the Canvas App. This combines two of the originally scoped agent concepts (a patient booking receptionist, and a symptom-to-department recommender) into a single cohesive conversational flow, rather than two disconnected agents — a symptom description leads naturally into a department suggestion, which leads naturally into booking, within one chat session.

This submission's agent operates over chat only. A telephone voice channel was scoped but not implemented, since it depends on a paid Azure telephony subscription — see Section 11.4.1 for the full reasoning. A third originally considered concept, allowing a doctor to conversationally hand off their schedule to a colleague, was also scoped but not built; it is documented in Section 11.4.6 as a direct extension of the Assign permission already configured on the Doctor security role (Section 5.4).

10.2 Architecture

The agent is built on Copilot Studio's Instructions-and-Tools architecture rather than the older Topics-only model, with the underlying conversational model connected to the CareTrack Dataverse environment directly through five configured Dataverse connector tools. The agent's Instructions define its persona (a professional, warm hospital receptionist), a symptom-to-department keyword mapping that always pairs a recommendation with a medical disclaimer, the full multi-step booking procedure, and an explicit constraint to answer only from connected tool data — never from general knowledge or web search — so the agent cannot invent hospital details, doctors, or availability that do not exist in Dataverse.

Tool	Dataverse Action	Purpose
Find Patient	List rows (Patients)	Checks for an existing patient, matching on phone number and name together — not phone number alone — since multiple patients can share a household phone number (Section 8.13).
Create Patient	Add a new row (Patients)	Creates a new Patient record when no existing match is found, capturing name, phone, email, and patient type.
List Doctors	List rows (Doctors)	Finds doctors matching the recommended or requested department/specialization.
List Doctor Schedules	List rows (Doctor Schedules)	Retrieves a doctor's weekly working hours so the agent can offer genuinely available slots.
Create Appointment	Add a new row (Appointments)	Writes the final Appointment record once patient, doctor, department, and a confirmed slot are all known, tagged with Booked Via = Copilot.

10.3 Symptom-to-Department Recommendation

When a patient describes a symptom in plain language (for example, “I have chest pain, what department should I see?”), the agent's Instructions map the described symptom to an appropriate department — Cardiology, in this example — and respond with that recommendation alongside an explicit disclaimer that this is not a medical diagnosis and a doctor should be consulted for accurate assessment. The agent then offers to continue directly into booking, asking for the patient's name, phone number, and email rather than requiring the patient to restart the conversation or navigate elsewhere.

This mapping lives entirely inside the agent's own Instructions rather than calling an external large language model API — a deliberate choice to keep the entire solution inside the Microsoft Power Platform ecosystem being demonstrated, and to avoid the latency and reliability risk of depending on a third-party service during a live demo. A faster, purpose-built classification model is identified as a future refinement in Section 11.4.2.

10.4 End-to-End Conversation Flow

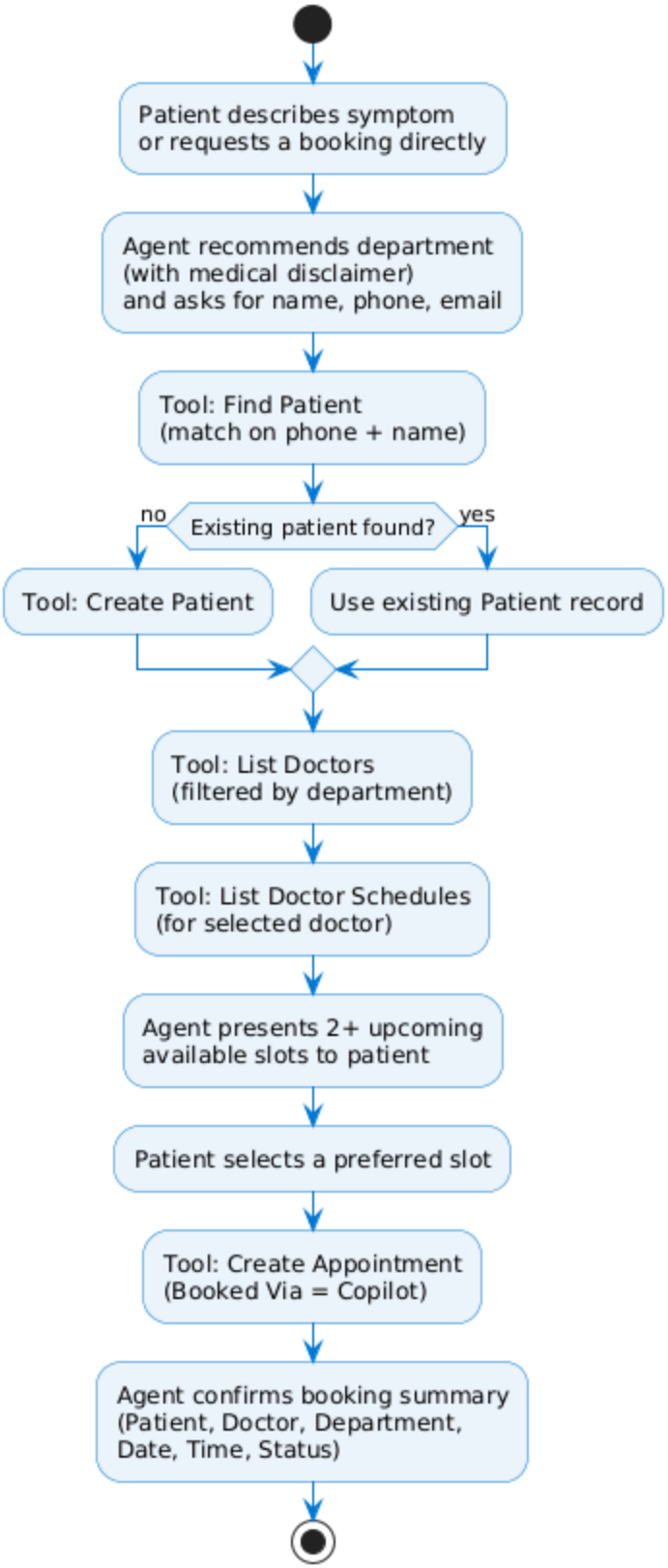


Fig 40. Copilot Agent End-to-End Conversation Activity Diagram

10.5 Tested End-to-End Example

The following exchange was run against the live agent and resulted in a real Appointment record being created in Dataverse, correctly linked to a newly created Patient record, an existing Doctor record, and the corresponding Hospital Department.

```
Patient: I have chest pain, what department should I see?

Agent: Based on your symptoms, I'd recommend seeing the Cardiology
department — they specialize in heart and chest-related concerns.

This is not a medical diagnosis — please consult a doctor for
accurate assessment.

Would you like me to book an appointment with a Cardiology doctor?
If so, I'll just need a few details: full name, phone number, email.

Patient: Sahr Ali, 123456789, hsahir327@gmail.com

Agent: [Tool: Find Patient → no existing match]
       [Tool: Create Patient → new record created]
       [Tool: List Doctors → found Dr. Amy, Cardiology]
       [Tool: List Doctor Schedules → found Monday availability]

Great news! Here's what I found:
Your Profile — Name: Sahr Ali, Phone: 123456789,
Email: hsahir327@gmail.com

Available Cardiology Appointment
Doctor: Dr. Amy | Specialization: Cardiology
Available Day: Mondays | Hours: 9:00 AM–5:00 PM
(Break: 12:00 PM–1:00 PM)

Here are 2 upcoming slots:
  1. Monday, June 22, 2026 — 9:00 AM
  2. Monday, June 22, 2026 — 2:00 PM

Which slot would you prefer?

Patient: option 1 is good to go

Agent: [Tool: Create Appointment]

Your appointment has been successfully booked! Summary:
Patient:    Sahr Ali
Doctor:     Dr. Amy
Department: Cardiology
Date:       Monday, June 22, 2026
Time:       9:00 AM
Status:     Scheduled
```

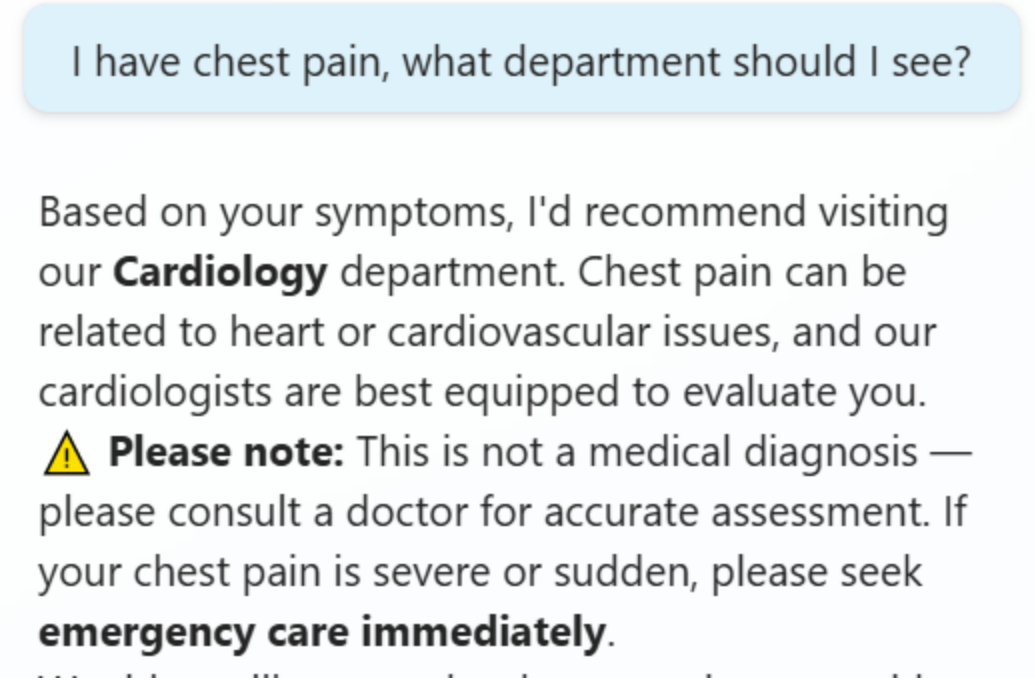


Fig 41. Prompt I have the issue we are facing

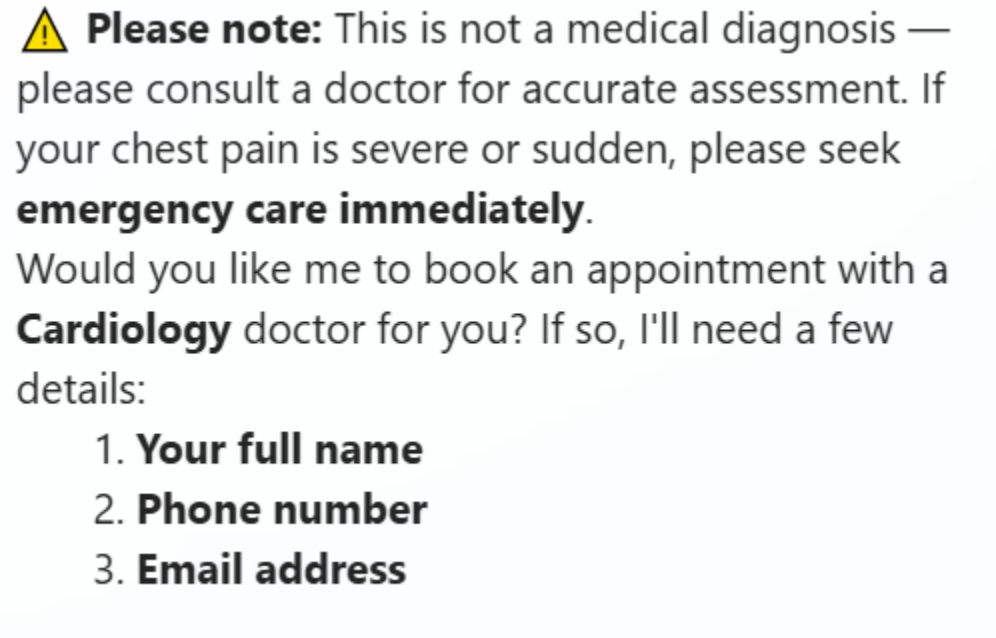


Fig 42. Caution as care for safety and don't the Agent as Dr, Provide Info

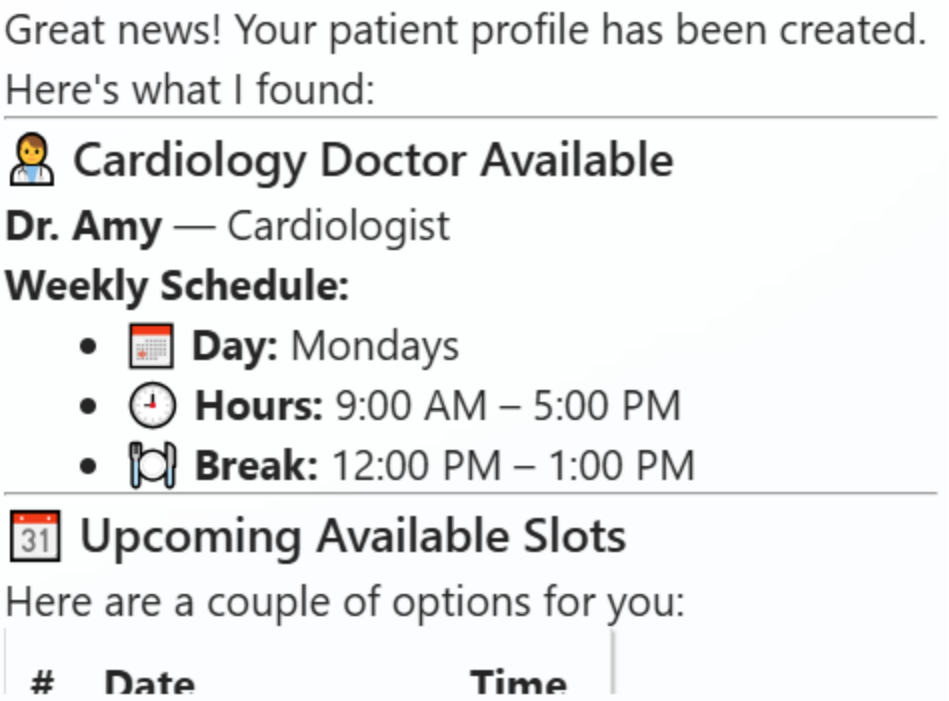


Fig 43. Available Slots with Department and Doctor

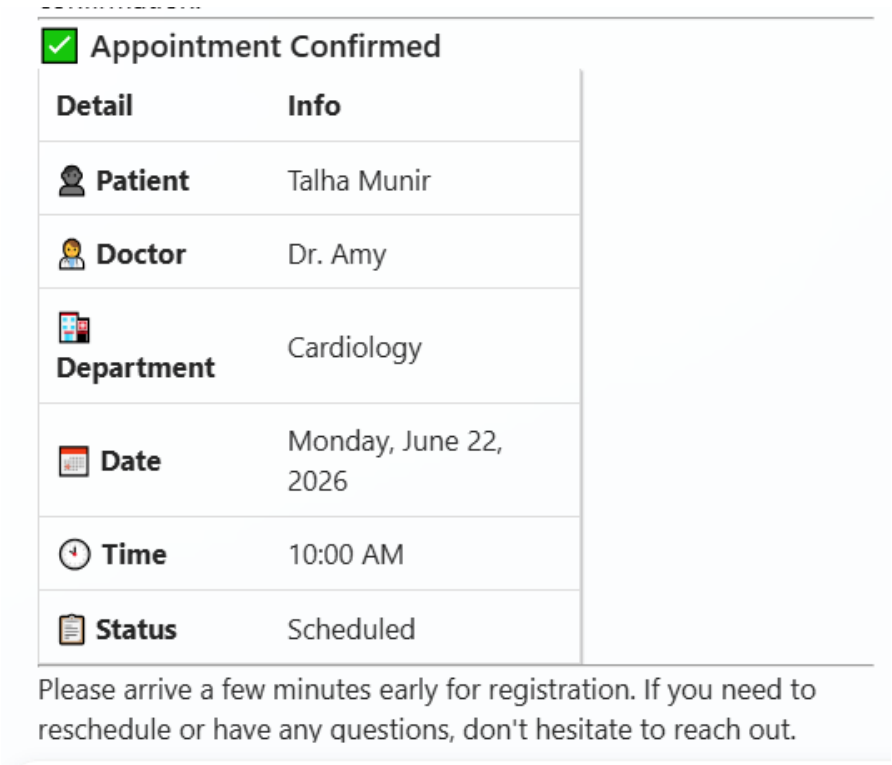


Fig 44. Provide Time and Appointment Confirmed

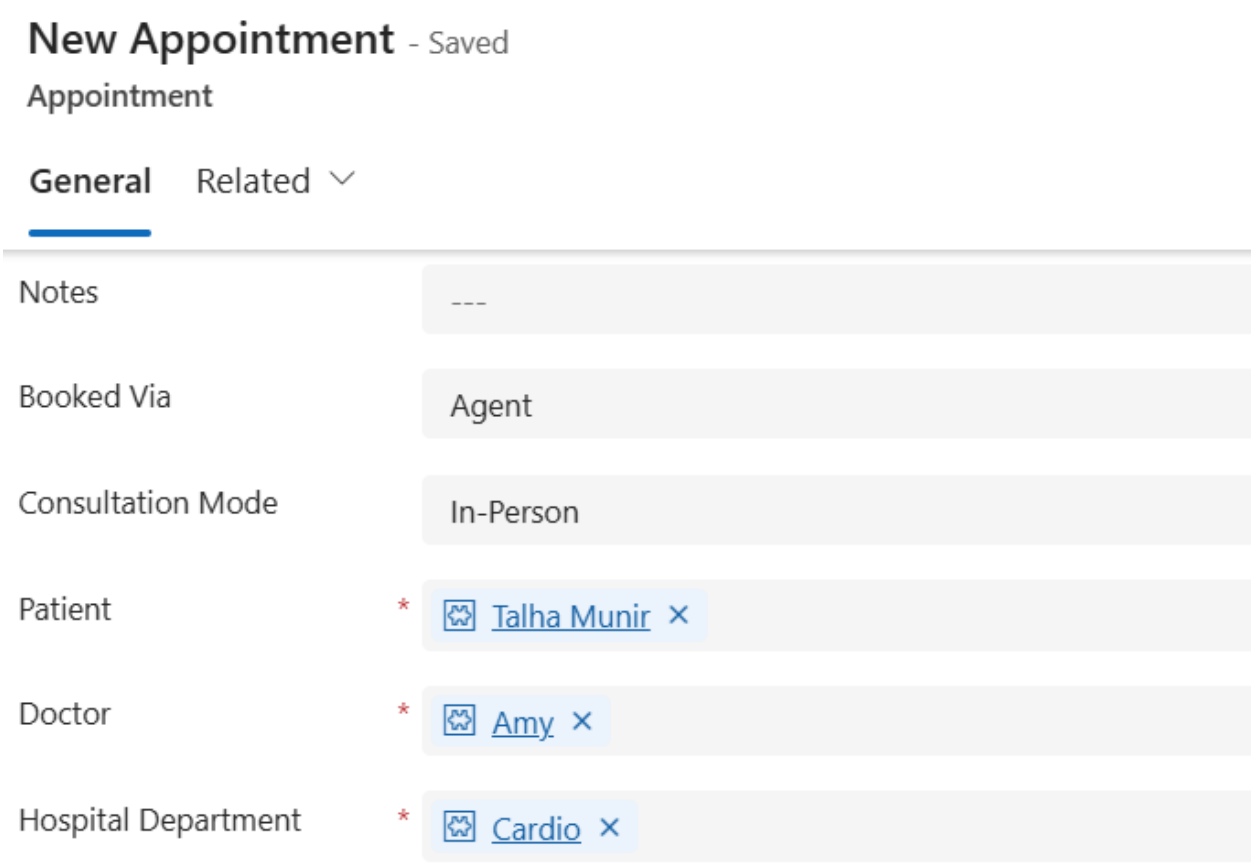


Fig 45. Appointment Added in Dataverse

10.6 Integration Issues Encountered and Resolved

Connecting Copilot Studio's Dataverse tools surfaced several integration-level errors during build-out, each rooted in OData syntax requirements that differ from how fields are referenced in Power Fx or the Model-Driven App designer. These are documented here as genuine troubleshooting evidence rather than omitted from the report.

Issue	Root Cause	Resolution
Find Patient returned HTTP 400: “Could not find a property named 'new_PatientName'”	OData field references are case-sensitive; the filter used the display-style casing instead of the column's logical name.	Corrected to the lowercase logical name new_patientname in the tool's filter configuration.
List Doctor Schedules returned HTTP 400: “Could not find a property named '_new_doctorid_value'”	Incorrect OData lookup-field reference — guessed rather than matched to the actual schema name.	Corrected to the real lookup reference, _new_doctor_value.
Create Appointment returned HTTP 400: “Resource not found for the segment '<guid>'”	Lookup fields (Doctor, Patient, Hospital Department) were passed as plain GUIDs;	Reformatted each lookup as /<entityset>(<guid>), e.g.

Issue	Root Cause	Resolution
	Dataverse's Add-a-new-row action requires OData bind (@odata.bind) syntax for lookups.	/new_doctors(<guid>), /new_patients(<guid>).
No Booked Via option represented agent-created bookings	The Booked Via choice column only had Walk-in, Canvas App, and Voice Agent options.	Added a new choice value, Copilot, and set it as a fixed input on the Create Appointment tool.

10.7 Status and Open Verification Item

The agent is fully built and was tested end-to-end as shown in Section 10.5, with a real Appointment record created and correctly linked to Patient, Doctor, and Hospital Department. One item remains to be explicitly re-verified following the timezone handling changes described in Section 8.14: that the Appointment Date Time written by the Create Appointment tool reflects the actual promised local slot (e.g., the literal “9:00 AM” offered to the patient) rather than being shifted by the same UTC storage offset that affected the Canvas App booking flow before correction.

11. Conclusion, Limitations, and Future Work

11.1 Summary of What Was Built

CareTrack demonstrates a complete, integrated Power Platform solution spanning a nine-table normalized Dataverse data model; six role-based security roles with three additional column-level security profiles; a Model-Driven App used by internal hospital staff, including a Business Process Flow visualizing the corporate account sales pipeline; a thirteen-screen Canvas App patient portal covering registration, multi-profile identity lookup, schedule-aware appointment booking, appointment history, billing visibility, OTP-based identity verification, and a self-service corporate enquiry funnel; a library of twenty-one Power Automate flows automating notifications, record creation, and operational alerts across the full patient and corporate account lifecycle; and a Copilot Studio chat-based conversational agent, combining symptom-to-department recommendation and end-to-end appointment booking into a single tool-driven assistant connected directly to Dataverse.

11.2 Design Principles Demonstrated

- Controlled, justified denormalization for reporting performance and historical accuracy, rather than blind third-normal-form adherence.
- Separation of legal/contractual state (Contract Start/End Date) from operational control (Is Active), allowing instant service suspension without altering legal records.
- A default-and-override data pattern (Corporate Account defaults, Employee-level overrides) minimizing data entry at scale.
- Scenario-driven security permissions (Assign/Share) granted only where a genuine clinical workflow justifies them, rather than uniform default configuration.
- Defense-in-depth security combining table-level RBAC with field-level column security masking.
- A business process, not just a data record, modeled for the corporate account lifecycle via a native Business Process Flow.
- An automation-first mindset, where state changes trigger their own downstream consequences (notifications, record creation, alerts) rather than depending on staff memory.

11.3 Known Limitations

- Doctor Schedule start/end/break times are stored as text rather than a native time type, since Dataverse does not offer a standalone time-only field — this is parsed at the application layer and is documented as a candidate for future refinement using a custom time-handling approach or a different storage strategy.
- Security role and column security masking could not be visually demonstrated live, since development used a single Administrator-level account that bypasses these restrictions by design; configurations were verified through the Dataverse security designer rather than live multi-user testing.

- The Employee Bulk Import flow (Flow 14) represents the most operationally complex automation in the system; Dataverse's native Excel Import feature on the Employee table offers a simpler alternative achieving the same outcome with less setup overhead.

11.4 Future Work

Several extensions were identified, scoped, and in some cases prototyped conceptually during development, but were deliberately not built into this submission — either because they depend on paid external services unsuitable for a student environment, or because they represent a natural next phase beyond the current project window rather than part of the agreed scope.

11.4.1 Telephone Voice Calling Agent (Azure Communication Services)

The CareTrack Booking Assistant described in Section 10 operates over chat only in this submission. Extending it to a genuine inbound telephone line — where a patient dials a real phone number and speaks to the agent exactly as they would a human receptionist — requires Azure Communication Services' telephony capability (a Direct Routing or Azure-provided phone number) connected to Copilot Studio's voice channel. This was identified as the natural next step toward a fully phone-accessible booking system, and the same Instructions and Tools already built and tested in Section 10 would carry over directly to a voice channel with no logic changes. It was not implemented in this submission because Azure Communication Services telephony is a paid, metered Azure service requiring a billing-enabled subscription, which was not available within the student/developer environment used for this project.

11.4.2 Faster, Lower-Latency Symptom Recommendation Model

The current symptom-to-department recommendation (Section 10.3) uses Copilot Studio's native topic and entity matching, chosen specifically to avoid external API dependencies during a live demo. A natural refinement for production use would be a purpose-built, lower-latency classification model fine-tuned specifically on a symptom-to-department mapping — reducing response time compared to a general-purpose conversational model and improving accuracy on ambiguous or multi-symptom descriptions. This was scoped as a future direction rather than built now, since it would require either a custom Azure AI model deployment or a paid third-party inference endpoint, both of which carry ongoing hosting costs beyond this project's scope.

11.4.3 Google Meet Links for Teleconsultations and HR Negotiation Calls

Flows 12 (HR Negotiation Meeting Scheduler) and 13 (Teleconsultation Meeting Link) are currently designed around Office 365 Outlook's native “Add online meeting” capability, which generates a Microsoft Teams link automatically as part of calendar event creation — the simplest path within the Microsoft connector ecosystem already used throughout the project. Generating Google Meet links instead would require integrating a Google Calendar connector alongside (or instead of) Office 365 Outlook, since Google Meet links are issued through Google Calendar's own event-creation API rather than Outlook's. This is a straightforward connector swap rather than a redesign of the underlying flow logic, and is noted as a configuration choice to revisit if the hospital's actual calendar infrastructure is Google Workspace rather than Microsoft 365.

11.4.4 Google Calendar Synchronization Flows

Beyond meeting links specifically, a broader future direction is two-way calendar synchronization — new flows that mirror confirmed Appointment records into a doctor's personal Google Calendar (or Outlook Calendar) automatically on booking, and conversely block out a doctor's Doctor Schedule availability if they mark personal leave directly in their calendar. This was scoped conceptually as a logical extension of the existing Doctor Schedule and Appointment automation but not built, since it introduces two-way sync conflict-resolution complexity (what happens if a doctor's calendar and Dataverse disagree) that was judged out of scope for the current submission.

11.4.5 Doctor-to-Doctor Schedule Handoff

A third agent concept considered alongside the booking assistant and symptom recommender was an internal, staff-facing capability allowing a doctor to conversationally request that their upcoming appointments be reassigned to a colleague — for example, ahead of planned leave. This was scoped but not built. It ties directly into the Assign permission already and deliberately granted to the Doctor security role on the Appointment table (Section 5.4), meaning the same underlying security design that supports manual reassignment in the Model-Driven App today would support conversational reassignment through the agent with no security-model changes — only a new Tool (a List rows + bulk Update against Appointments, filtered by doctor and date) and a confirmation step before any records are changed, consistent with how Create Appointment and Create Patient already require an explicit patient confirmation before writing data (Section 10.2).

11.4.6 Other Identified Extensions

- Migrate Doctor Schedule time fields to a more robust time-handling strategy as Dataverse's native field types evolve, removing the need for text-based minute-math parsing described in Section 8.14.
- Introduce multi-user testing across all six security roles with dedicated non-administrative test accounts, to validate table-level and column-level restrictions under realistic conditions rather than configuration-only verification.
- Add Power BI dashboards for executive-level visibility into appointment volume, corporate account health, and revenue by billing channel, consuming the same Dataverse tables documented in Section 4.
- Revisit the Employee Bulk Import flow (Flow 14) against Dataverse's native Excel Import feature, which may offer a simpler path to the same outcome with less ongoing maintenance overhead than a dedicated Power Automate flow.
- Add a patient-facing notification preference center, allowing patients to choose which of the email/SMS notification flows they wish to receive, rather than all flows defaulting to email.

CareTrack illustrates that a student project built on Microsoft Power Platform can move beyond a simple CRUD demonstration to model a genuine, multi-stakeholder business

relationship — hospital, employer, and patient — with the operational rules, automation, and security architecture that a real production system would require.

WE HOPE YOU LIKE OUR PROJECT FOR WHICH WE HAVE WORKED EFFORTLESSLY AS THIS MIGHT BE THE LAST PROJECT WE MIGHT BE COMPLETING.